

#1 · Alignment

What is camber?

Answer

Camber is the inward or outward tilt of the top of the tire, viewed from the front of the vehicle. Positive camber means the top of the tire leans out, away from the vehicle. Negative camber means the top leans in, toward the vehicle.

#2 · Alignment

What tire wear pattern does excessive camber cause?

Answer

Excessive camber wears one shoulder of the tread smooth while the rest of the tread stays healthy. Too much positive camber wears the outside shoulder, and too much negative camber wears the inside shoulder. This smooth, single-shoulder wear is different from the saw-tooth feathering caused by toe error.

#3 · Alignment

What is the cross-camber pull rule?

Answer

A vehicle pulls toward the side that has more positive camber, since a tilted tire rolls like a cone and steers toward the direction it leans. If the left front reads more positive camber than the right, the vehicle pulls left. Manufacturers set a maximum allowed cross-camber, often around 0.5 degrees, to prevent this pull.

#4 · Alignment

How is camber adjusted on a short-long-arm (SLA) suspension versus a MacPherson strut?

Answer

SLA suspensions often use shims at the upper control arm mounting points to change camber. MacPherson strut vehicles typically use an eccentric camber bolt at the strut-to-knuckle connection, elongated mounting holes, or an aftermarket camber bolt kit. Some modern vehicles have no factory camber adjustment at all.

#5 · Alignment

Why does a rear-wheel-drive vehicle sometimes get slightly more positive camber on the left side on purpose?

Answer

Most roads slope down to the right for drainage, which is called road crown. Setting slightly more positive camber on the left offsets the vehicle's tendency to drift right on a crowned road. This is a deliberate factory setting, not an error.

#6 · Alignment

What is caster?

Answer

Caster is the forward or rearward tilt of the steering axis, viewed from the side of the vehicle. The steering axis runs through the upper and lower steering pivot points. Positive caster means the top of the steering axis tilts rearward.

#7 · Alignment

What benefit does positive caster give a vehicle?

Answer

Positive caster gives directional stability at speed and helps the steering wheel return to center after a turn, similar to how a shopping cart caster trails behind its own pivot. More positive caster improves stability and returnability but increases steering effort. Power steering allowed manufacturers to specify much more positive caster than older manual-steering vehicles used.

#8 · Alignment

Does caster cause tire wear?

Answer

No, caster is not a tire-wearing angle. A vehicle can have badly unequal caster side to side and show no unusual tread wear at all. On the exam, shoulder wear points to camber and feathered wear points to toe, while caster problems show up as handling complaints instead.

#9 · Alignment

What is the cross-caster pull rule?

Answer

A vehicle pulls toward the side with less positive caster, since the side with more positive caster is more stable and resists turning. This is the opposite direction rule from camber, where pull goes toward more positive camber. Technicians sometimes correct a mild pull by cross-adjusting caster on a vehicle with no camber adjustment.

#10 · Alignment

What symptoms result from too little caster?

Answer

Too little caster causes wander, light steering effort, and poor return to center after a turn. The steering axis is not tilted back enough to create a strong self-centering effect. This often shows up as the driver needing to make constant small steering corrections.

#11 · Alignment

What symptoms result from too much caster?

Answer

Too much caster causes hard steering effort and snap-back after turns, and it can amplify road shocks felt through the wheel. The steering axis is tilted back farther than specified, increasing the self-centering force beyond the intended amount. This is the opposite problem from insufficient caster.

#12 · Alignment

What is toe?

Answer

Toe describes whether the front wheels point toward or away from each other, viewed from above. Toe-in, also called positive toe, means the fronts of the tires are closer together than the rears. Toe-out, or negative toe, means the fronts are farther apart than the rears.

#13 · Alignment

What is the difference between total toe and individual toe?

Answer

Total toe is the combined toe reading of both wheels added together. Individual toe is each single wheel's angle measured relative to the vehicle centerline or thrust line. Toe can be expressed in degrees, millimetres, or inches.

#14 · Alignment

What tire wear pattern does toe error cause?

Answer

Toe error causes feathered, saw-tooth wear, where each tread rib develops a sharp edge on one side and a rounded edge on the other. This happens because toe error scrubs the tire sideways with every rotation. A customer complaint of rapid wear with a saw-tooth feel points directly to toe.

#15 · Alignment

Why is toe always set last in an alignment?

Answer

Adjusting camber or caster moves the steering knuckle and steering arm, which changes the toe reading. If toe were set first, those later camber and caster adjustments would undo it. So the correct order is camber and caster first, then toe.

#16 · Alignment

Why do rear-wheel-drive and front-wheel-drive vehicles use different static toe specifications?

Answer

Rear-wheel-drive vehicles are usually set with slight toe-in, since rolling resistance pushes the front wheels rearward and outward while driving, and the toe-in compensates so the wheels run at zero toe on the road. Front-wheel-drive vehicles are often set at zero or slight toe-out, since drive torque pulls the front wheels forward and inward under power.

#17 · Alignment

Where is front toe adjusted?

Answer

Front toe is adjusted at the tie rods, either by turning adjuster sleeves on a parallelogram linkage or by turning the inner tie rod or its jam-nut adjuster on a rack-and-pinion system. Both sides must be adjusted equally to keep the steering wheel centered. The wheel is held straight ahead with a steering wheel holder during the adjustment.

#18 · Alignment

What is steering axis inclination (SAI)?

Answer

SAI is the inward tilt of the steering axis, viewed from the front of the vehicle. Like caster, it angles the steering axis toward the tire contact patch, which reduces scrub radius and creates a self-centering effect at low speed. SAI is a built-in angle on virtually all vehicles.

#19 · Alignment

Is steering axis inclination (SAI) adjustable?

Answer

No, SAI is a non-adjustable angle built into the shape of the spindle and knuckle. This makes it a strong diagnostic tool, because if SAI is wrong, something structural is bent. It is not something a technician sets during a normal alignment.

#20 · Alignment

What is included angle (IA) and how is it calculated?

Answer

Included angle is the angle between the steering axis and the centerline of the wheel, fixed by the shape of the spindle and knuckle. The formula is $IA = SAI + \text{camber}$, adding camber as a signed number so negative camber subtracts. IA is a fixed, non-adjustable angle.

#21 · Alignment

If SAI reads 8 degrees and camber reads negative 1 degree, what is the included angle?

Answer

Included angle equals SAI plus camber, so 8 plus negative 1 equals 7 degrees. This shows how a negative camber reading lowers the included angle value. Always add camber as a signed number rather than always adding it as a positive value.

#22 · Alignment

How do you use SAI and included angle to diagnose a bent part?

Answer

If camber is out of specification but included angle matches the specification and the other side, the spindle and knuckle are straight and the problem is inboard, such as a bent strut, bent control arm, or shifted cradle. If included angle itself is wrong on one side, the spindle or knuckle is bent. Remember that IA points at the knuckle and spindle, while SAI points at the strut, tower, or cradle.

#23 · Alignment

What is scrub radius?

Answer

Scrub radius is the distance at the road surface between the tire contact patch centerline and the point where the steering axis meets the ground. Positive scrub radius means the steering axis meets the ground inboard of the tire centerline. Negative scrub radius means it meets the ground outboard of the tire centerline.

#24 · Alignment

What effect does a larger positive scrub radius have?

Answer

A larger positive scrub radius increases steering effort and magnifies the pull caused by a dragging brake or an underinflated tire. It also increases kickback felt through the steering wheel over bumps. Scrub radius is designed into the vehicle and not directly adjustable.

#25 · Alignment

Why do many front-wheel-drive vehicles use negative scrub radius on purpose?

Answer

Negative scrub radius helps the vehicle track straight if one front brake circuit fails or one front tire blows out. In that failure, negative scrub radius causes the affected wheel to toe slightly against the imbalance instead of pulling hard to one side. This is a deliberate safety-related design choice.

#26 · Alignment

What can accidentally change a vehicle's scrub radius?

Answer

Installing wheels with a different offset, mounting oversized tires, or adding wheel spacers all move the tire centerline outward or inward. This changes the scrub radius even though the suspension geometry itself was not touched. Technicians should be aware that aftermarket wheel changes can affect steering feel this way.

#27 · Alignment

What is setback?

Answer

Setback is the condition where one front wheel sits farther back than the other, so the front axle line is not perpendicular to the vehicle centerline. A small amount of setback is a normal manufacturing tolerance. Significant setback usually points to collision damage or a shifted structural component.

#28 · Alignment

What commonly causes significant setback?

Answer

Significant setback is usually caused by collision damage, a bent strut rod or lower control arm, or a shifted crossmember. It is a sign that a structural or suspension component has moved out of its normal position. Small setback within tolerance is not a concern.

#29 · Alignment

What symptom does significant setback produce, and is it adjustable?

Answer

Setback effectively shortens the wheelbase on one side and can cause the vehicle to lead or pull toward the setback side. Setback is a diagnostic measurement only, not something a technician adjusts. Finding it points to a bent part that needs repair or replacement.

#30 · Alignment

What is the thrust angle?

Answer

The thrust angle is the angle between the thrust line and the geometric centerline of the vehicle. The geometric centerline runs through the midpoints of the front and rear axles. The thrust line is the direction the rear wheels actually push the vehicle, based on rear total toe.

#31 · Alignment

When is the thrust angle equal to zero?

Answer

The thrust angle is zero when the rear individual toe readings are equal on both sides. In that case the thrust line matches the geometric centerline exactly. Any imbalance in rear toe side to side creates a nonzero thrust angle.

#32 · Alignment

What does a significant thrust angle cause?

Answer

A significant thrust angle makes the vehicle dog-track, meaning the rear wheels push the body slightly sideways. The driver corrects with the steering wheel, so the vehicle travels crabbed down the road with the steering wheel off center, even if front toe is set perfectly. Rear tire wear and an uncenterable steering wheel are the classic clues.

#33 · Alignment

What is the difference between a two-wheel alignment and a thrust-line alignment?

Answer

A two-wheel, or front-end, alignment sets only the front angles referenced to the geometric centerline and assumes the rear axle is straight. A thrust-line alignment instead measures the rear thrust angle and sets the front wheels parallel to the thrust line, centering the steering wheel even if rear toe cannot be adjusted. Thrust-line alignment is the minimum acceptable method for solid rear axle vehicles.

#34 · Alignment

What happens during a four-wheel alignment, and why is the order rear-then-front?

Answer

A four-wheel alignment corrects rear toe and camber first so the thrust line matches the geometric centerline, then sets the front angles to that corrected reference. The rear is done first because it establishes the reference line the front wheels are aligned to. This is required practice on any vehicle with adjustable rear suspension, including nearly all front-wheel-drive cars.

#35 · Alignment

What is toe-out on turns, and what causes it?

Answer

Toe-out on turns describes how the inside front wheel steers to a greater angle than the outside front wheel during a turn, since the inside wheel travels a tighter circle. It is produced by Ackermann steering geometry, where the steering arms angle inward toward the center of the rear axle. It is not adjustable, since it is built into the shape of the steering arms.

#36 · Alignment

In a typical toe-out on turns specification, if the outside wheel is steered to 20 degrees, what might the inside wheel read?

Answer

A typical specification might call for the inside wheel to read about 22 degrees when the outside wheel is at 20 degrees. The inside wheel always steers to a greater angle because it travels a tighter circle during the turn. This difference is checked with turn plates during a full alignment.

#37 · Alignment

What causes incorrect toe-out on turns, and what symptom does it produce?

Answer

Incorrect toe-out on turns is almost always caused by a bent steering arm, and the fix is replacing the bent part rather than adjusting anything. A vehicle with this problem squeals its tires during slow, tight turns like parking maneuvers, even though its static toe setting is correct. Tire squeal in parking lot turns combined with correct static toe points directly to a bent steering arm.

#38 · Alignment

How do you convert 0.5 degrees into minutes?

Answer

One degree contains 60 minutes, so you multiply the decimal fraction by 60 to convert to minutes. For 0.5 degrees, 0.5 times 60 equals 30 minutes. So 0.5 degrees equals 30 minutes.

#39 · Alignment

What does it mean if a vehicle has no factory camber adjustment and camber reads out of specification?

Answer

On vehicles with no built-in camber adjustment, out-of-spec camber points to bent or worn suspension parts rather than a lost adjustment. Common causes include a bent strut, control arm, or spindle. Replacing the damaged part is the repair, not forcing an adjustment that does not exist.

#40 · Alignment

What hardware do technicians use to adjust caster?

Answer

Caster adjustment methods mirror camber adjustment methods: shims at the control arm, eccentric cams, strut rods with adjustable length, or slotted frames that let a component be repositioned and retorqued. Many strut-equipped vehicles offer no caster adjustment at all. When no adjustment exists, incorrect caster points to a bent or damaged part instead.

#41 · Alignment

In what units can toe be expressed?

Answer

Toe can be expressed in degrees, in millimetres, or in inches, and a technician needs to recognize all three formats on the exam. A toe reading may be given in any of these units depending on the alignment machine or specification sheet. Regardless of the unit, toe-in always means the fronts of the tires are closer together than the rears.

#42 · Alignment

How does SAI relate to scrub radius?

Answer

Steering axis inclination angles the steering axis toward the tire contact patch, which reduces scrub radius. A smaller scrub radius means less distance between the tire's contact patch and the point where the steering axis meets the ground. This reduces steering effort and lessens road shock felt through the steering wheel.

#43 · Alignment

What does zero scrub radius mean?

Answer

Zero scrub radius means the steering axis meets the ground at the exact same point as the tire contact patch centerline. This differs from positive scrub radius, where the steering axis meets the ground inboard of the tire centerline, and negative scrub radius, where it meets the ground outboard. Scrub radius is designed into the vehicle and is not something a technician adjusts directly.

#44 · Alignment

What is the geometric centerline of a vehicle?

Answer

The geometric centerline is an imaginary line drawn through the midpoints of the front and rear axles. It serves as the vehicle's true center reference line for alignment purposes. The thrust angle is measured as the angle between this line and the rear thrust line.

#45 · Alignment

What rear condition brings the thrust line back in line with the geometric centerline?

Answer

The thrust line matches the geometric centerline when rear individual toe is equal on both sides of the vehicle. A four-wheel alignment corrects rear toe and camber specifically to achieve this. Once the thrust line and geometric centerline coincide, the thrust angle is zero.

#46 · Alignment

How do you convert 1 degree 15 minutes into decimal degrees?

Answer

To convert minutes into decimal degrees, divide the number of minutes by 60. Here 15 minutes divided by 60 equals 0.25, so 1 degree 15 minutes equals 1.25 degrees. This reverse conversion is used whenever a specification is written in degrees and minutes instead of decimal form.

#47 · Alignment

Why must ride height be checked before any alignment angle is measured?

Answer

Camber, caster, and toe all change as the suspension moves through its travel, so ride height sets the baseline those angles are measured from. A sagging spring or torsion bar lowers the vehicle and skews every reading. If ride height is out of specification, the alignment numbers are meaningless until the suspension fault is corrected first.

#48 · Alignment

A vehicle shows camber and toe both out of specification on one side. What should the technician check first?

Answer

Ride height, not the angles themselves. Ride height is the prerequisite for every alignment angle, so it must be measured and corrected before adjusting camber or toe. Jumping straight to angle adjustment on a vehicle with wrong ride height produces readings that will not hold.

#49 · Alignment

How is ride height typically adjusted on a torsion bar suspension versus a coil spring suspension?

Answer

Torsion bar vehicles allow ride height adjustment at the anchor bolt, so a sagging side can often be corrected by turning the bolt. Coil spring and leaf spring vehicles usually have no such adjustment and require spring replacement instead. This is why torsion bar height is checked and adjusted early in the pre-alignment inspection.

#50 · Alignment

What is the correct way to check a load-carrying ball joint before an alignment?

Answer

The joint must first be unloaded with a jack placed to relieve spring pressure off it. Once unloaded, axial and radial play are measured and compared against the manufacturer's specification. Checking it while still loaded can hide real looseness and let a worn joint pass inspection.

#51 · Alignment

How does checking a follower ball joint differ from checking a load-carrying ball joint?

Answer

A follower joint is checked simply for any perceptible looseness, without needing to unload spring pressure first. A load-carrying joint instead must be unloaded and its play measured against a numeric specification. Confusing the two methods can produce a false pass on a load-carrying joint.

#52 · Alignment

List the main components inspected in a pre-alignment inspection besides ball joints and ride height.

Answer

The technician checks tire pressure and matched tire size and wear, wheel bearings for looseness, tie rod ends, idler and pitman arms, control arm bushings, strut mounts, and stabilizer links. Any looseness found must be corrected before the alignment proceeds. Loose parts make every alignment reading unrepeatable.

#53 · Alignment

Why must loose steering and suspension components be repaired before performing an alignment?

Answer

Play in a joint or bushing makes every camber, caster, or toe reading unrepeatable, since the wheel can shift position under the sensor's own weight or a light push. The alignment would appear to succeed on the machine but the readings would not hold on the road. Fixing looseness first is what makes the final numbers trustworthy.

#54 · Alignment

What vehicle loading condition does most alignment specifications call for before measuring angles?

Answer

Most specifications call for a full fuel tank or an equivalent ballast weight added to the vehicle. Some manufacturers also specify particular passenger or cargo weights to be placed in specific seats or the trunk. Loading the vehicle to spec matters because suspension geometry and ride height change with vehicle weight.

#55 · Alignment

Where are the front and rear tires positioned when a vehicle is driven onto an alignment rack?

Answer

The front tires are centered on the turn plates and the rear tires sit on slip plates. This positioning lets the front wheels pivot freely for the caster sweep and turning radius checks. The slip plates let the rear wheels move slightly without binding as the suspension settles.

#56 · Alignment

What is runout compensation and why is it performed first on the alignment machine?

Answer

Runout compensation rotates each wheel, or rolls the vehicle on rolling-compensation systems, so the machine can measure and cancel out error caused by a bent rim or a clamp that was not mounted perfectly true. Skipping this step bakes the clamp or rim error directly into the camber and toe readings. It must happen before any angle is trusted.

#57 · Alignment

What happens if a technician skips wheel runout compensation before measuring alignment angles?

Answer

Any error from a bent rim or a mounting clamp that is not perfectly true gets baked directly into the camber and toe readings. The machine has no way to distinguish real angle error from clamp or rim error without this step. The result is inaccurate readings that can send the technician chasing the wrong adjustment.

#58 · Alignment

What is the caster sweep and why is it necessary?

Answer

Caster is the one primary alignment angle the machine cannot read statically, so the wheels are steered through a prescribed arc, typically 10 to 20 degrees each way, with the brake pedal locked by a depressor. The machine calculates caster and steering axis inclination from how camber changes through that arc. Without the sweep, caster and SAI could not be measured at all.

#59 · Alignment

Why is the brake pedal locked with a depressor during the caster sweep?

Answer

Locking the pedal keeps the wheels from rolling forward or backward while they are steered through the sweep arc. This keeps the sweep measurement isolated to the steering motion needed to calculate caster and SAI. Without a locked pedal, wheel rotation could introduce error into the sweep reading.

#60 · Alignment

What is the correct order of adjusting camber, caster, and toe, and why?

Answer

Camber and caster are adjusted first, and toe is always adjusted last. Changing camber or caster moves the steering arm and knuckle position, which in turn changes toe. Adjusting toe before camber and caster would simply be undone by the later adjustments.

#61 · Alignment

Why is the steering wheel locked level before the final toe adjustment?

Answer

Locking the steering wheel level ensures the finished vehicle drives straight with a centered steering wheel once toe is set. If toe were adjusted with the wheel off center, the vehicle could track straight but the wheel would sit crooked to the driver. This final step ties the toe setting to a properly centered wheel.

#62 · Alignment

How many minutes are in one degree of an alignment specification, and how do you convert 0.25 degrees to minutes?

Answer

One degree contains 60 minutes. To convert a decimal degree fraction to minutes, multiply it by 60, so 0.25 degrees equals 15 minutes. This conversion skill is needed because specs and machine printouts can be given in either decimal degrees or degrees and minutes.

#63 · Alignment

Convert 1.75 degrees into degrees and minutes.

Answer

Multiply the decimal portion, 0.75, by 60 to get 45 minutes. So 1.75 degrees equals 1 degree 45 minutes. The whole-degree portion stays the same and only the decimal fraction converts.

#64 · Alignment

Convert 1 degree 15 minutes into decimal degrees.

Answer

Divide the minutes portion by 60, so 15 divided by 60 equals 0.25. Add that to the whole degree to get 1.25 degrees. This is the reverse of converting decimal degrees into minutes.

#65 · Alignment

What is a cross tolerance on an alignment spec sheet, and why can a vehicle fail it even when both individual readings pass?

Answer

A cross tolerance limits how much camber or caster can differ between the left and right side, separate from each side's individual specification range. Two readings can each fall within their own individual tolerance and still exceed the allowed left-right difference. A vehicle that fails cross tolerance will pull even though neither side is individually out of spec.

#66 · Alignment

On an SLA suspension using shims at the upper control arm shaft, what is the difference between adjusting camber and adjusting caster?

Answer

Adding or removing an equal number of shims at both mounting bolts changes camber. Adding or removing a different number of shims at one bolt versus the other changes caster instead. The same shim hardware serves both angles depending on how it is applied.

#67 · Alignment

What are eccentric cams used for in alignment adjustment, and on what type of suspension are they common?

Answer

Eccentric cams are mounted at control arm pivot points and rotate to swing the arm inward or outward, changing camber or caster. They are common on both SLA and some strut suspensions as a built-in adjustment method. Rotating the cam repositions the arm without needing separate shims.

#68 · Alignment

What adjustment method do many front-wheel-drive cars use to correct non-adjustable rear camber and toe?

Answer

Many front-wheel-drive cars accept rear camber and toe shims, which are tapered plates installed between the rear spindle and its mounting surface. These shims correct geometry that otherwise has no built-in adjustment. They let a rear angle be brought into specification without replacing the spindle or knuckle.

#69 · Alignment

What is the correct repair when an alignment angle is out of specification but the vehicle has no adjustment for that angle?

Answer

The correct repair is to replace the bent or sagged component causing the error, such as a strut, control arm, or spring. Modern vehicles should never have components forced or bent to create an adjustment that does not exist. Forcing parts risks hidden stress damage and an unsafe repair.

#70 · Alignment

How is front toe physically adjusted on a rack-and-pinion steering system versus a parallelogram linkage?

Answer

On a rack-and-pinion system, toe is adjusted by turning the inner tie rod or its jam-nut adjuster. On a parallelogram linkage, toe is adjusted by turning adjuster sleeves along the tie rods. In both systems, both sides must be adjusted equally to keep the steering wheel centered.

#71 · Alignment

Why must both tie rods be adjusted equally when setting front toe?

Answer

Adjusting both sides equally keeps the steering wheel centered once toe is set to specification. If only one side is turned, the wheel spokes could end up off center even though total toe reads correctly. A steering wheel holder is used during the adjustment to keep the wheels locked straight ahead while this happens.

#72 · Alignment

What is a two-wheel (front-end) alignment and when is it acceptable to perform?

Answer

A two-wheel alignment sets only the front angles, referenced to the geometric centerline, and assumes the rear axle is true without measuring it. It is the oldest alignment method on the exam and is only acceptable when the rear axle is confirmed to be straight. It cannot correct for any rear thrust angle.

#73 · Alignment

What is a thrust-line alignment and why is it the minimum standard for solid rear axle vehicles?

Answer

A thrust-line alignment measures the rear thrust angle and sets the front wheels parallel to the thrust line instead of the geometric centerline. This centers the steering wheel even when the rear toe on a solid axle cannot be adjusted. It is the minimum acceptable method whenever the rear axle is fixed and non-adjustable.

#74 · Alignment

What is a four-wheel alignment and on what type of vehicles is it required?

Answer

A four-wheel alignment adjusts the rear wheels first, bringing rear toe and camber into specification so the thrust line matches the geometric centerline, then sets the front angles to that corrected reference. It is required on any vehicle with adjustable rear suspension, which includes nearly all front-wheel-drive cars. The rear is always set before the front in this method.

#75 · Alignment

In a four-wheel alignment, why is the rear axle adjusted before the front?

Answer

The rear defines the thrust line reference that the front wheels are aligned to, so it must be corrected first. Setting the front before the rear would mean aligning to a reference that is about to change. Once the rear thrust line matches the geometric centerline, the front angles can be set accurately to it.

#76 · Alignment

A customer's rear tires show wear and the steering wheel cannot be centered even though front toe is set perfectly. What should the technician suspect?

Answer

A significant rear thrust angle is the likely cause. The rear wheels are pushing the vehicle slightly sideways, forcing the driver to hold the steering wheel off center to travel straight. Front toe adjustment alone cannot fix a problem that originates at the rear axle.

#77 · Alignment

What does toe-out on turns measure, and how is it checked?

Answer

Toe-out on turns describes how much more the inside wheel must steer compared to the outside wheel during a turn, produced by the Ackermann geometry of the steering arms. It is checked using turn plates during a full alignment, comparing the inside wheel angle against the outside wheel angle at a specified setting. It is a diagnostic check, not an adjustable angle.

#78 · Alignment

A vehicle squeals its tires during slow, tight parking lot turns, but its static toe measures within specification. What is the likely cause?

Answer

The likely cause is incorrect toe-out on turns, usually from a bent steering arm. Since toe-out on turns only shows up during sharp turning and is not adjustable, correct static toe readings do not rule it out. Turn plates must be used to check it, and the fix is replacing the bent steering arm.

#79 · Alignment

A vehicle pulls steadily to one side while driving straight. What are the most likely causes to check?

Answer

The most likely causes are cross-camber, cross-caster, a defective tire, or a dragging brake. Swapping the front tires side to side is a simple way to test whether the tire itself is causing the pull. Road crown can also cause a mild version of the same symptom.

#80 · Alignment

A driver reports having to constantly make small corrections to hold a lane, without a strong pull to either side. What does this symptom point to?

Answer

This wander symptom points to insufficient caster, excessive play in the steering linkage, or loose wheel bearings. Unlike a steady pull, wander is not about a difference side to side but a general lack of directional stability. Checking caster specification and linkage looseness is the logical next step.

#81 · Alignment

The steering wheel sits off center but the vehicle tracks straight down the road. What two causes should be considered?

Answer

This points to either front toe having been set unevenly between the two sides, or the rear thrust angle having been ignored during the alignment. Both cause the wheel spokes to sit crooked while the vehicle itself still travels straight. Correcting equal front toe or accounting for thrust line normally fixes it.

#82 · Alignment

A vehicle steers poorly back to center after completing a turn. What conditions commonly cause poor returnability?

Answer

Poor returnability points to low caster, a binding ball joint, or a binding strut bearing. Caster is what gives a vehicle its self-centering effect after a turn, so insufficient caster directly weakens that return. A physically binding component can produce the same symptom even with correct caster.

#83 · Alignment

What is memory steer and what usually causes it?

Answer

Memory steer is when the vehicle continues steering in the direction of the last turn instead of returning to straight ahead. It is almost always caused by a binding upper strut mount or a binding ball joint, rather than an alignment angle being out of specification. The fix is freeing or replacing the binding component, not adjusting angles.

#84 · Alignment

What is torque steer and what commonly causes it?

Answer

Torque steer is a pull that occurs during hard acceleration on front-wheel-drive vehicles. It comes from unequal drive axle lengths or angles, or from a ride height difference side to side, rather than from a toe error. Because it only appears under acceleration, it is diagnosed differently from a steady alignment-related pull.

#85 · Alignment

What is bump steer and what typically causes it?

Answer

Bump steer is a steering twitch felt when one front wheel hits a bump in the road. It indicates a tie rod that is not level with its control arm, which can come from a bent steering arm, incorrect ride height, or an improperly installed steering rack. It is a geometry problem rather than a simple angle being out of tolerance.

#86 · Alignment

Why must a steering angle sensor be calibrated or reset after an alignment that changes toe?

Answer

The steering angle sensor must know exactly where straight ahead is for electronic stability control and ADAS systems to function correctly. Changing toe or repositioning the steering wheel shifts that reference point. Calibration is performed with a scan tool or through the manufacturer's self-learn drive procedure so the sensor's zero point matches the newly aligned wheels.

#87 · Alignment

Besides the steering angle sensor, what other calibration may be required after an alignment that changes the rear thrust line?

Answer

Forward camera and radar calibration may also be required on vehicles equipped with lane keeping assist or adaptive cruise control. These systems rely on knowing the vehicle's true straight-ahead direction, which can shift when the thrust line changes. Skipping this calibration leaves those safety systems reading from an outdated reference.

#88 · Alignment

Why does the 310E exam treat ADAS calibration as part of a complete alignment rather than an optional extra?

Answer

Returning an aligned vehicle without required calibrations leaves safety systems like stability control and lane keeping working from false data about where straight ahead actually is. An alignment changes the very reference points those systems depend on. The exam expects the calibration step to be included whenever toe or thrust line is changed, not treated as separate or optional work.

#89 · Alignment

What alignment machine step calculates caster and steering axis inclination together, and why can neither be measured statically?

Answer

The caster sweep calculates both caster and steering axis inclination by measuring how camber changes as the wheels are steered through a prescribed arc. Unlike camber and toe, caster has no fixed static reading the machine can capture with the wheels held straight ahead. Steering axis inclination is derived from that same sweep motion rather than measured directly.

#90 · Alignment

What must be verified about all four tires during the pre-alignment inspection, and why does it matter?

Answer

All four tires must be checked to confirm they match in size and wear, and tire pressures must be set to the vehicle's placard specification. Mismatched tire sizes or uneven wear can throw off ride height and handling independent of the alignment angles themselves. Correcting tire condition first prevents chasing a pull or wear complaint that is really coming from the tires.

#91 · Alignment

Why is a full alignment inspection considered wasted work if performed over worn suspension parts?

Answer

Worn ball joints, tie rod ends, or bushings introduce play that shifts under the machine's own measurement process, so the readings will not repeat or hold once the vehicle is driven. Any adjustment made against those readings is unreliable and can come back out of specification quickly. That is why the documented pre-alignment inspection of components always comes before angle measurement begins.

#92 · Alignment

During the pre-alignment inspection, what should be done if a wheel bearing is found to be loose?

Answer

The loose wheel bearing must be adjusted or replaced as needed before proceeding with the alignment. A loose bearing lets the wheel shift position independent of the suspension geometry, corrupting any angle reading taken while it is loose. Correcting it first ensures the final alignment readings will actually hold on the road.

#93 · Braking Systems

What does Pascal's law state about a confined fluid?

Answer

Pressure applied anywhere in a confined fluid is transmitted equally and undiminished in all directions throughout that fluid. This is why pressing the brake pedal sends the same line pressure to every wheel at the same instant. It is the basic physical principle behind every hydraulic brake system.

#94 · Braking Systems

How do you calculate hydraulic pressure in a brake circuit?

Answer

Pressure equals force divided by the area the force acts on. In brakes this is usually stated in kilopascals or megapascals. A small master cylinder piston pushed hard creates high pressure, and that same pressure then acts on every piston downstream.

#95 · Braking Systems

Why does a larger caliper piston produce more clamping force than a smaller wheel cylinder piston?

Answer

Since line pressure is the same everywhere in the circuit, output force depends only on piston area. A bigger piston has more surface area for the pressure to push against, so it produces more force. The trade off is that the bigger piston moves a shorter distance, since fluid volume is conserved.

#96 · Braking Systems

What is the main job of the master cylinder?

Answer

The master cylinder converts the driver's pedal force into hydraulic fluid pressure. It has one or more pistons that pressurize fluid inside a bore and push it out to the wheel cylinders and calipers. It is the starting point of the entire hydraulic circuit.

#97 · Braking Systems

Why do modern master cylinders use a dual circuit (tandem) design?

Answer

A tandem master cylinder has two pistons in one bore, each feeding a separate hydraulic circuit. If one circuit loses fluid, the other circuit still works so the vehicle can still stop. Pedal travel gets longer and stopping distance increases, but total brake failure is avoided.

#98 · Braking Systems

How do the primary and secondary pistons work in a tandem master cylinder?

Answer

The primary piston is pushed directly by the pushrod, and it pushes the secondary piston through fluid pressure and a spring. Each piston feeds its own circuit. If the primary circuit fails, the primary piston bottoms out and pushes the secondary piston mechanically, and if the secondary circuit fails, the secondary piston bottoms against the end of the bore while the primary circuit still works.

#99 · Braking Systems

What is a front/rear split hydraulic system and where is it used?

Answer

A front/rear split puts both front brakes on one circuit and both rear brakes on the other circuit. Since the front brakes do most of the braking work, losing the front circuit leaves weak stopping power. This split is common on rear-wheel-drive vehicles with fairly even weight distribution.

#100 · Braking Systems

What is a diagonal split hydraulic system and why is it used on front-wheel-drive cars?

Answer

A diagonal split pairs the left front brake with the right rear brake, and the right front with the left rear, on two separate circuits. If either circuit fails, one front brake still works, keeping roughly half of total braking capability. It pairs with negative scrub radius steering geometry so the car does not pull hard during a partial system failure.

#101 · Braking Systems

What do the primary cup seal and secondary seal do inside a master cylinder piston?

Answer

The primary cup seal builds and holds hydraulic pressure as the piston moves forward. The secondary seal sits behind the piston and keeps fluid from leaking out the back of the bore. Both seals must stay in good condition or the master cylinder will bypass fluid internally.

#102 · Braking Systems

What does the vent (compensating) port do in a master cylinder?

Answer

The vent port connects the reservoir to the pressure chamber when the piston is fully retracted at rest. It lets fluid expand and contract with temperature changes and allows the system to compensate as pads and shoes wear. If this port stays covered, trapped fluid can expand and cause brake drag.

#103 · Braking Systems

What does the replenishing port do in a master cylinder?

Answer

The replenishing port keeps the area behind the primary cup seal full of fluid. This prevents the cup from starving for fluid when the piston releases quickly. Without it, rapid pedal release could draw air into the system.

#104 · Braking Systems

What happens if a master cylinder pushrod is adjusted too long?

Answer

A pushrod that is too long holds the piston slightly forward at rest, so the cup never uncovers the vent port. Trapped fluid then expands as the brakes heat up during driving. This causes the brakes to drag and can even lock the wheels.

#105 · Braking Systems

What is a quick take-up master cylinder designed to do?

Answer

It uses a stepped bore where a large rear section moves a large volume of fluid at low pressure to quickly take up pad clearance. A quick take-up valve then bypasses fluid so the smaller front section can build high working pressure. This design is used with low-drag calipers that retract their pistons farther than normal.

#106 · Braking Systems

What symptom points to a failed quick take-up valve?

Answer

A failed quick take-up valve shows up as excessive pedal travel on the very first brake application. This happens because the large low-pressure volume is not being bypassed properly. Low-drag calipers rely on this valve to avoid a long first stroke.

#107 · Braking Systems

What is the chemical difference between DOT 3/4/5.1 fluid and DOT 5 fluid?

Answer

DOT 3, DOT 4, and DOT 5.1 are polyglycol based fluids and can be mixed with each other in an emergency. DOT 5 is silicone based, is dyed purple, and does not absorb water. Silicone fluid must never be mixed with polyglycol fluid.

#108 · Braking Systems

Why is DOT 5 silicone fluid unsuitable for anti-lock brake systems?

Answer

DOT 5 fluid aerates easily, meaning it traps air bubbles more readily than polyglycol fluid. Since it does not absorb water, any moisture that enters the system pools as free droplets at low points instead of dispersing. Those droplets can boil or freeze right where they sit, which is a problem inside precise ABS hydraulic units.

#109 · Braking Systems

What happens if you mix DOT 3, DOT 4, and DOT 5.1 fluids together?

Answer

These three fluids are all polyglycol based and are compatible for emergency mixing. However, mixing them lowers the overall performance of the blend toward the weakest fluid in the mix. This means the boiling point of the mixed fluid drops toward the lowest rated fluid used.

#110 · Braking Systems

What is the difference between dry and wet boiling points for brake fluid?

Answer

The dry boiling point applies to fresh, unused fluid straight from a sealed container. The wet boiling point applies to fluid that already contains about 3.7 percent water, which happens naturally over time. Wet boiling points are always lower, which is why old fluid fades sooner under hard braking.

#111 · Braking Systems

What does it mean that brake fluid is hygroscopic?

Answer

Hygroscopic means the fluid absorbs moisture directly from the surrounding air. This happens gradually through hoses, seals, and the reservoir vent over time. Absorbed water lowers the fluid's boiling point and corrodes steel lines, cylinders, and ABS components.

#112 · Braking Systems

What causes brake fluid fade and how does it feel to the driver?

Answer

Fluid fade happens when wet, moisture contaminated fluid heats up during hard braking and boils, forming vapor bubbles in the line. Vapor compresses easily, unlike liquid fluid, so pedal pressure gets absorbed instead of transmitted. The driver feels the pedal sink toward the floor with reduced stopping power.

#113 · Braking Systems

How often should brake fluid typically be flushed, and why?

Answer

Most manufacturers recommend flushing brake fluid every two to three years, though the published service schedule always governs. Fluid absorbs moisture over time even in a sealed system, which lowers boiling point and promotes corrosion. Regular flushing removes that moisture before it causes fade or internal damage.

#114 · Braking Systems

What do copper test strips check for in brake fluid?

Answer

Copper test strips measure the amount of dissolved copper in the fluid, which is a corrosion by-product from moisture attacking metal parts. About 200 parts per million of copper is commonly cited as the point where fluid should be replaced. This test is one of three common ways technicians evaluate fluid condition, along with boiling point testers and moisture meters.

#115 · Braking Systems

What material are rigid brake lines made from?

Answer

Rigid brake lines are made of double-walled steel tubing, usually coated for corrosion resistance. Copper-nickel alloy line is also approved in most jurisdictions because it bends and flares easily. Both materials must withstand high hydraulic pressure without cracking.

#116 · Braking Systems

Why is plain copper tubing never acceptable for brake lines?

Answer

Plain copper tubing work-hardens over time from vibration and pressure cycling. Once it work-hardens it becomes brittle and can crack under normal hydraulic pressure. This makes it unsafe for any brake line application.

#117 · Braking Systems

What is the difference between an SAE double flare and an ISO bubble flare?

Answer

The SAE double (inverted) flare is a 45 degree flare folded back on itself, common on many domestic vehicles. The ISO bubble flare is a rounded button shape common on European and many newer domestic vehicles. The two styles are not interchangeable, and mismatching them to the wrong seat will leak under pressure.

#118 · Braking Systems

Why is a single flare never acceptable on a brake line?

Answer

A single flare is a simple cone shape without the folded-back reinforcement of a double or bubble flare. Under the high pressure of a brake system, a single flare will split open. Only double (SAE) or bubble (ISO) flares are approved for brake line connections.

#119 · Braking Systems

What symptom does an internally restricted brake hose cause?

Answer

A restricted hose acts like a one-way check valve, letting pressure pass to the caliper but not release. That wheel then drags and pulls to one side, often becoming worse only after the brakes warm up. A collapsed inner liner inside the hose can cause the same kind of pull.

#120 · Braking Systems

Can a damaged brake hose be repaired?

Answer

No, a damaged flexible brake hose must always be replaced, never repaired. Any hose that got twisted during installation must be reinstalled correctly or also replaced. Hoses should be inspected for cracking, bulging, chafing, twisting, and a soft or spongy feel.

#121 · Braking Systems

Why should a flare-nut (line) wrench be used on brake line fittings?

Answer

A flare-nut wrench wraps almost all the way around the fitting instead of just gripping two flats like an open-end wrench. This spreads force evenly and keeps the soft brass or steel nut from rounding off. Rounded fittings become very difficult to remove or tighten properly later.

#122 · Braking Systems

What does a proportioning valve do?

Answer

A proportioning valve limits the rate of pressure rise to the rear brakes once pressure passes a set split point. This keeps the lightly loaded rear wheels from locking up during hard braking. Some trucks use a height-sensing version linked to the rear axle so a loaded truck gets more rear pressure than an empty one.

#123 · Braking Systems

What does the metering (hold-off) valve do on a front-disc, rear-drum system?

Answer

The metering valve delays fluid pressure to the front disc calipers until modest pressure has already built up. This gives the rear drum shoes time to overcome their return springs before the front brakes engage. The result is that both axles apply braking force together, even during light stops.

#124 · Braking Systems

What three components does a combination valve package together?

Answer

A combination valve packages the metering valve, the proportioning valve, and a pressure differential switch into one housing. This saves space and simplifies plumbing compared to three separate valves. Each part still performs its own separate hydraulic function.

#125 · Braking Systems

How does the pressure differential switch turn on the brake warning lamp?

Answer

The switch compares hydraulic pressure between the two circuits inside the combination valve. If one circuit loses pressure, the internal piston shifts off-center and grounds the switch, lighting the red warning lamp. On some designs the piston can stay off-center after a repair, so the lamp stays on until bleeding re-centers it.

#126 · Braking Systems

What pressure does a residual pressure check valve hold, and why?

Answer

A residual pressure check valve holds roughly 40 to 80 kilopascals, about 6 to 12 psi, of standing pressure in drum brake circuits. This small pressure keeps the wheel cylinder cup lips sealed against their bores. It also helps keep air from being drawn past the cups when the brakes are released.

#127 · Braking Systems

Why do disc brake circuits not use a drum-style residual pressure valve?

Answer

Disc brake calipers sit very close to the rotor with minimal clearance by design. Even a small amount of standing residual pressure would be enough to keep the pads dragging on the rotor. Drum brakes need the residual pressure to seal their cups, but disc brakes do not, so the valve is left out of disc circuits.

#128 · Braking Systems

How does manual bleeding of the hydraulic system work?

Answer

Manual bleeding uses a helper who pumps and holds the pedal down while the technician opens and closes each bleeder screw in turn. The pedal must never be released while a bleeder screw is open, or air will be drawn back into the system. This method removes trapped air that causes a low, spongy pedal.

#129 · Braking Systems

How does pressure bleeding work and what tool does it often require on front-disc, rear-drum systems?

Answer

Pressure bleeding attaches a tank of fluid under about 100 to 140 kilopascals to an adapter on the reservoir, pushing fresh fluid through each bleeder screw. This lets one technician do the whole job alone. Many front-disc, rear-drum systems need a metering valve override tool, since the metering valve would otherwise block low-pressure flow to the front.

#130 · Braking Systems

How does vacuum bleeding work?

Answer

Vacuum bleeding pulls fluid out through the bleeder screw using a hand pump or a shop vacuum tool. Small bubbles drawn in past the bleeder screw threads are considered normal with this method and do not always mean trapped air. It is a common one-person bleeding method in the shop.

#131 · Braking Systems

How does gravity bleeding work and when is it useful?

Answer

Gravity bleeding simply opens each bleeder screw and lets the reservoir's own head pressure slowly push fluid through the lines. It is slow but very gentle on the system. It is often used after a component replacement, sometimes combined with another bleeding method to finish the job.

#132 · Braking Systems

What is the traditional wheel bleeding sequence on a front/rear split system?

Answer

The traditional sequence starts at the wheel farthest from the master cylinder and works toward the closest one. On a front/rear split this is commonly right rear, then left rear, then right front, then left front. Diagonally split and ABS-equipped vehicles often specify a different order, so the manufacturer's published sequence should always be followed.

#133 · Braking Systems

Why must a new or rebuilt master cylinder be bench bled before installation?

Answer

Bench bleeding strokes the master cylinder's pistons with the outlet lines looped back into the reservoir before it goes on the vehicle. This removes air trapped inside the bore from manufacturing or storage. Skipping this step can make bleeding the system on the vehicle nearly impossible afterward.

#134 · Braking Systems

What is the self-energizing effect in a drum brake?

Answer

Self-energizing happens when drum rotation tries to drag the leading shoe around with it as the shoe contacts the spinning drum. This wedges the shoe harder into the drum than hydraulic pressure alone would produce. It is the basic friction effect that makes drum brakes powerful for their size.

#135 · Braking Systems

How does a leading-trailing (non-servo) drum brake design work?

Answer

In a leading-trailing design, each shoe anchors independently at the bottom of the backing plate. The leading (forward) shoe self-energizes while the trailing shoe does not, so braking force is uneven between the two shoes. The design brakes about equally well in both directions, which suits the rear axle of front-wheel-drive vehicles.

#136 · Braking Systems

How does a duo-servo drum brake design work?

Answer

In a duo-servo design, the two shoes are linked at the bottom by an adjuster, with only the top anchor fixed in place. The primary shoe's self-energizing force transfers through that adjuster link into the secondary shoe, called servo action. The secondary shoe does most of the work, so it uses longer or thicker lining than the primary shoe.

#137 · Braking Systems

What happens if duo-servo drum brake shoes are installed in swapped positions?

Answer

The primary and secondary shoes in a duo-servo brake are different, with the secondary shoe carrying more lining because it does more work. Swapping their positions sends the servo force through the wrong shoe. This causes grabby, uneven braking at that wheel.

#138 · Braking Systems

How do drum brake self-adjusters keep lining clearance correct over time?

Answer

Self-adjusters take up clearance as the lining wears down, keeping pedal travel consistent. Duo-servo brakes usually adjust through a star wheel operated by a cable or lever mechanism that indexes when the brakes are applied while backing up. Many leading-trailing brakes instead adjust whenever the parking brake is applied.

#139 · Braking Systems

What symptom points to a seized drum brake self-adjuster?

Answer

A seized adjuster typically shows up as a low brake pedal that improves after the driver pumps it a few times. It is usually paired with a low, soft-feeling parking brake as well. Both symptoms happen because the shoes never move closer to the drum as lining wears.

#140 · Braking Systems

How is a brake drum measured, and what are the two size limits that matter?

Answer

A drum is measured with a drum micrometer at several points around its friction surface. The machine-to limit is the largest diameter a worn drum may be machined to while still leaving wear allowance before it reaches the discard diameter. The discard (maximum) diameter is the larger limit stamped or cast on the drum, and any drum at or beyond it must be replaced, not machined.

#141 · Braking Systems

What is the difference between a fixed caliper and a floating (sliding) caliper?

Answer

A fixed caliper bolts rigidly to the knuckle and has pistons on both sides of the rotor that clamp simultaneously, making it stiff and powerful but heavier and costlier. A floating caliper has pistons on the inboard side only, and the caliper body slides on pins to pull the outboard pad into the rotor. Seized or dry slide pins on a floating caliper are a classic cause of uneven pad wear and drag.

#142 · Braking Systems

Why do disc brakes generally resist fade and pull better than drum brakes?

Answer

Disc brakes shed heat far better than enclosed drums because the rotor and pads are more exposed to airflow. They are also self-cleaning in water since the open design lets water sling off easily. Discs have no self-energizing action, so they need higher hydraulic pressure and power assist, but that same lack of self-energizing makes their performance more consistent and predictable.

#143 · Braking Systems

What is a suspended vacuum design in a vacuum booster?

Answer

It is the layout used on virtually all modern boosters where both the front and rear chambers see engine vacuum at rest. Since pressure is equal on both sides of the diaphragm, the diaphragm does not move and no assist is applied. Assist only begins when the pedal opens the rear chamber to atmospheric air.

#144 · Braking Systems

How does pressing the brake pedal create assist inside a vacuum booster?

Answer

The pushrod moves an air control valve that first closes the passage between the two chambers, then opens the rear chamber to filtered outside air. Atmospheric pressure now pushes on the rear of the diaphragm while vacuum stays in the front chamber. This pressure difference drives the diaphragm forward and adds force to the master cylinder pushrod.

#145 · Braking Systems

What happens to a vacuum booster at run-out?

Answer

Run-out is the point where full atmospheric pressure has already reached the rear chamber of the booster. No more assist can be produced no matter how much harder the pedal is pushed. Any extra stopping force beyond this point must come from the driver's own leg.

#146 · Braking Systems

Why do manufacturers use a dual (tandem) diaphragm booster instead of a single diaphragm?

Answer

Stacking two smaller diaphragms on one shaft adds their working areas together. This produces the assist force of a much larger single diaphragm booster while fitting in a smaller housing. It is done for extra assist in tight spaces, not for backup redundancy.

#147 · Braking Systems

Why do many diesel and gasoline direct injection engines need a separate vacuum pump for the brake booster?

Answer

These engines produce very little usable manifold vacuum during normal running. An engine-driven or electric vacuum pump supplies the booster instead of relying on intake manifold vacuum. Without this pump the booster would have little or no assist available.

#148 · Braking Systems

What does the vacuum check valve at the booster do?

Answer

It is a one-way valve that holds the highest vacuum level the booster has reached. It stops that vacuum bleeding back to the manifold when manifold vacuum drops, such as at wide open throttle. It also preserves reserve vacuum for one or two assisted stops after the engine stalls.

#149 · Braking Systems

Describe the booster function test.

Answer

With the engine off, pump the pedal several times to use up any stored vacuum. Hold light pressure on the pedal and start the engine. The pedal should drop slightly under your foot as vacuum builds and assist begins, and no drop means no assist is present.

#150 · Braking Systems

What is the booster leak-down or vacuum retention test?

Answer

Run the engine to charge the booster fully, then shut it off and wait several minutes. The booster should still provide one or two power-assisted brake applications from the vacuum it stored. If assist disappears immediately, the booster, check valve, or hose is leaking.

#151 · Braking Systems

What happens if the booster output pushrod is adjusted too long?

Answer

A pushrod that is too long holds the master cylinder primary piston slightly applied at rest. This blocks the compensating (vent) port, so fluid trapped in the system cannot expand normally. As the brakes heat up, trapped fluid expands, pressure builds, and the brakes can drag and eventually lock.

#152 · Braking Systems

What does hydro-boost use as its power source instead of engine vacuum?

Answer

Hydro-boost uses hydraulic pressure supplied by the power steering pump. A spool valve inside the unit restricts return flow and directs pump pressure against a power piston. This multiplies pedal force and sends it on to the master cylinder.

#153 · Braking Systems

Why can a failing power steering pump affect brake pedal feel on a hydro-boost vehicle?

Answer

The same pump that supplies the power steering gear also supplies the hydro-boost unit. A weak pump, slipping drive belt, or low fluid level reduces pressure available to both systems at once. This is why a hydro-boost complaint should always start with checking the power steering system.

#154 · Braking Systems

What does the accumulator in a hydro-boost unit provide?

Answer

It is a spring-loaded or gas-charged reservoir that stores high pressure fluid from the pump. If the engine stalls or the drive belt breaks, the accumulator supplies one to three assisted brake applications. After it depletes, the pedal becomes very hard but the base hydraulic brakes still function.

#155 · Braking Systems

What does bleeding a hydro-boost system actually involve?

Answer

It means bleeding the power steering circuit, not the brake fluid inside the booster itself. The steering wheel is turned lock to lock with the engine off, then again with the engine running while applying the brake pedal. Foamy or milky fluid means air is still trapped and the process must be repeated.

#156 · Braking Systems

What is an electro-hydraulic or electromechanical brake booster?

Answer

It is a power assist system, common on hybrids and electric vehicles, that uses an electric motor driving a ball screw or plunger, or an electric pump charging an accumulator, to generate braking pressure on demand. A pedal simulator gives the driver a normal pedal feel while a control module decides actual wheel pressure. This lets the system blend friction braking with regenerative braking.

#157 · Braking Systems

Before opening the hydraulic system on an electro-hydraulic brake unit, what must a technician do first?

Answer

The technician must follow the manufacturer's procedure to release stored accumulator pressure before disconnecting any lines. Many of these units also require scan tool service modes before service begins. Skipping this step can leave dangerously high pressure trapped in the lines.

#158 · Braking Systems

What is the purpose of an antilock braking system (ABS)?

Answer

ABS prevents wheel lockup during hard braking so the driver keeps steering control. A locked wheel has almost no lateral grip, so ABS holds each wheel near its peak slip range instead of letting it skid. This shortens stopping distance on most surfaces while keeping the vehicle steerable.

#159 · Braking Systems

Name the three core hardware components of an ABS system.

Answer

The system uses wheel speed sensors at each wheel, a hydraulic control unit (HCU) containing solenoid valves and a pump motor, and an electronic control module. The module is usually built onto the HCU as one assembly. Together they monitor wheel speed and adjust brake pressure automatically.

#160 · Braking Systems

How does a passive wheel speed sensor generate its signal?

Answer

A passive sensor is a permanent magnet wrapped with a coil of wire, mounted near a toothed tone ring on the wheel or axle. As the tone ring teeth pass the sensor tip, they induce an AC sine wave voltage. The frequency and amplitude of this signal rise as wheel speed increases.

#161 · Braking Systems

How does an active wheel speed sensor differ from a passive one?

Answer

An active sensor uses a Hall-effect or magnetoresistive element and needs a supply voltage from the module. It outputs a digital square wave signal whose frequency tracks wheel speed. Unlike a passive sensor, it reads accurately down to almost zero speed and can often detect wheel direction.

#162 · Braking Systems

Why can't a passive wheel speed sensor read very low speeds accurately?

Answer

The signal from a passive sensor comes from the tone ring inducing voltage as it spins past the coil. As wheel speed drops toward zero, the signal amplitude falls toward zero as well, making the signal too weak to read reliably. Active sensors do not have this limitation because they are externally powered.

#163 · Braking Systems

What do the inlet (isolation) and outlet (dump) valves do inside an ABS hydraulic control unit?

Answer

The normally open inlet valve controls pressure flowing to the wheel from the master cylinder, and closing it holds pressure steady. The normally closed outlet valve opens to dump fluid into a low pressure accumulator, dropping pressure so a locking wheel can spin back up. The pump motor then returns this fluid to the circuit, which is what causes pedal pulsation.

#164 · Braking Systems

List the three stages of an ABS control cycle.

Answer

Hold closes the inlet valve so pressure at that wheel cannot rise further. Release (dump) opens the outlet valve to let fluid escape and pressure drop so the wheel can recover. Apply closes the outlet valve and reopens the inlet valve, often in pulses, to rebuild pressure.

#165 · Braking Systems

What is select-low logic in a three-channel ABS system?

Answer

A three-channel system controls each front wheel individually but treats both rear wheels as a single channel. Select-low logic means the rear wheel closest to lockup sets the pressure applied to both rear wheels together. This keeps the rear axle stable even though the two rear wheels are not controlled separately.

#166 · Braking Systems

What is RWAL (rear-wheel antilock)?

Answer

RWAL, also called RABS, is an older single-channel system used mainly on older pickups that controls only the rear axle through one valve assembly. It typically uses a single speed sensor mounted on the differential or transmission. It only prevents rear wheel lockup under changing box loads; the front wheels can still lock up.

#167 · Braking Systems

What does the ABS bulb check at startup show?

Answer

On key on, the module runs a self-test and briefly lights the amber ABS warning lamp. If everything checks out, the lamp turns off after this brief self-test. An amber lamp that stays on afterward means antilock function is disabled and a code is stored.

#168 · Braking Systems

What is the difference between the amber ABS lamp and the red BRAKE lamp staying on?

Answer

The amber ABS lamp means antilock and related electronic functions are disabled, but base brakes still work normally. The red BRAKE lamp points to a base brake problem such as low fluid, the park brake applied, or loss of a hydraulic circuit. If both lamps are on together, the base hydraulic problem should be treated as the priority.

#169 · Braking Systems

How do you test a passive wheel speed sensor with a DMM and a lab scope?

Answer

A DMM checks coil resistance against the specification to find most circuit faults, and reading AC voltage while spinning the wheel by hand confirms basic output. A healthy sensor typically produces a few hundred millivolts AC at hand-spin speed, rising with speed. The definitive test is a lab scope, which should show a clean, even sine wave with no dropouts.

#170 · Braking Systems

How is an active wheel speed sensor tested differently from a passive one?

Answer

An active sensor cannot be meaningfully resistance-tested with a DMM the way a passive sensor can. Instead the technician verifies supply voltage at the connector and checks the switched digital signal with a scope. Watching live wheel speed data on the scan tool while rotating the wheel also confirms proper operation.

#171 · Braking Systems

What causes a wheel speed sensor air gap to grow too wide, and what problem does that cause?

Answer

A wide air gap commonly comes from wheel bearing wear, a sensor that is not fully seated, or rust jacking under a bolt-on sensor bracket. A wide gap weakens a passive sensor's signal strength. This causes low speed signal dropouts and can trigger false diagnostic codes.

#172 · Braking Systems

How does a corroded or damaged tone ring cause false ABS activation?

Answer

A cracked or rust-swollen tone ring produces a repeating glitch in the sensor waveform once per wheel revolution. The module can misread this glitch as a wheel rapidly approaching lockup during normal light braking. The classic symptom is false ABS cycling, pedal pulsation, and a ratcheting noise at low speed during a routine stop.

#173 · Braking Systems

When is a scan tool actuated bleed needed on an ABS-equipped vehicle?

Answer

It is needed when air has entered the hydraulic control unit itself, for example after the reservoir ran dry or the HCU was replaced. Trapped air hides behind closed solenoid valves so the pedal stays spongy no matter how long conventional bleeding continues. The scan tool commands the module to cycle the valves and run the pump so air can be pushed out while bleeding each wheel in sequence.

#174 · Braking Systems

What is traction control (TCS) and how does it slow a spinning drive wheel?

Answer

Traction control is essentially ABS logic working during acceleration instead of braking. When a drive wheel spins faster than the others, the module can apply that wheel's brake through the hydraulic control unit and can also ask the engine controller to reduce torque. Braking the spinning wheel transfers torque across an open differential to the wheel that still has grip.

#175 · Braking Systems

Why does brake-based traction control help so much on a split-friction surface, such as one wheel on ice?

Answer

An open differential normally sends torque to whichever wheel has the least resistance, which is the slipping wheel on ice. Braking that spinning wheel adds resistance to it, so torque is redirected across the differential to the wheel that still has grip. This lets the vehicle move forward using the wheel with traction instead of just spinning the icy wheel.

#176 · Braking Systems

What three extra sensors does electronic stability control (ESC) use beyond standard ABS sensors?

Answer

ESC adds a steering angle sensor on the column that tells the module where the driver is pointing the wheels. A yaw rate sensor measures how fast the vehicle body is rotating about its vertical axis. A lateral accelerometer measures the sideways force acting on the vehicle.

#177 · Braking Systems

How does ESC respond to an understeer condition where the vehicle plows wide of the intended path?

Answer

ESC compares what the driver is asking for through steering input against what the vehicle is actually doing. In understeer, the module can brake the inside rear wheel to help rotate the nose of the car back into the turn. This individual wheel braking effectively acts as an extra steering tool.

#178 · Braking Systems

How does ESC respond to an oversteer condition where the rear of the vehicle steps out?

Answer

The module detects that the rear of the vehicle is rotating faster than the driver's steering input calls for. It brakes the outside front wheel to help pull the car straight again. It can also reduce engine torque as needed to help regain stability.

#179 · Braking Systems

Why does a steering angle sensor often need recalibration after an alignment or steering component replacement?

Answer

The steering angle sensor tells the stability control module exactly where the driver has pointed the front wheels. If it is not zeroed with a scan tool after alignment work, a battery disconnect, or steering component replacement, its reading can be off. An uncalibrated sensor can trigger false ESC intervention or an unwanted warning lamp.

#180 · Braking Systems

What does electronic brake force distribution (EBD) replace, and how does it work?

Answer

EBD replaces the mechanical proportioning valve used on older vehicles. Using the same wheel speed sensors and ABS hydraulic valves, the module continuously limits rear brake pressure so the rear wheels approach lockup at the same rate as the fronts. It adjusts automatically for vehicle load, deceleration rate, and road surface.

#181 · Braking Systems

Why might a vehicle with an ABS fault also lose EBD function?

Answer

EBD works through the same hydraulic control unit valves that ABS uses to control wheel pressure. If an ABS fault disables those valves, rear pressure limiting through EBD is lost along with antilock function. Some vehicles light the red brake lamp along with the amber ABS lamp for exactly this reason.

#182 · Braking Systems

On a rear disc drum-in-hat parking brake, how does the mechanism work?

Answer

A small dedicated drum brake is built into the center hat section of the rear rotor. Small shoes expand inside this hat section when the parking brake cable is pulled, holding the vehicle. Adjustment is made through a star wheel accessed through a hole plus cable adjustment at the equalizer.

#183 · Braking Systems

Why should a technician never press an integrated parking brake caliper piston straight in with a C-clamp?

Answer

An integrated caliper uses the parking brake cable to rotate a lever and internal screw mechanism that pushes the piston out mechanically. Pressing the piston straight in with a C-clamp instead of rotating it back damages this internal adjuster mechanism. The piston on this design must be rotated while being pressed back during pad service.

#184 · Braking Systems

What must a technician do before servicing rear pads on a motor-on-caliper electric park brake (EPB)?

Answer

The system must first be placed into service or maintenance mode using a scan tool or the manufacturer's specified procedure. This allows the motor to fully retract the piston mechanism before the pads are removed. Skipping this step risks damaging the actuator or leaves the piston unable to retract properly.

#185 • Braking Systems

Why does an electric park brake usually need a scan tool relearn after pad replacement?

Answer

After new pads are installed, the system needs to recalibrate so it can set proper clamp force and correct pad position. Without this relearn, the electric actuator may not apply the correct holding force. The relearn is typically performed with a scan tool following the manufacturer's procedure.

#186 • Braking Systems

Besides lighting the brake lamps, what other functions does the stop lamp switch control?

Answer

The stop lamp switch signal tells the ABS and stability control modules that the driver is braking. It also releases the shift interlock so the vehicle can be shifted out of park, and it cancels cruise control. A misadjusted or failed switch can cause brake light problems, a shifter stuck in park, or ABS and cruise control complaints.

#187 • Braking Systems

How does an electric pad wear sensor warn the driver that pads are worn out?

Answer

An electric wear sensor embeds a wire loop inside the brake pad material. On most designs, the rotor grinds through this loop as the pad wears down, opening the circuit and triggering the warning light. Electric wear sensors are single-use and get replaced along with the pads once worn through.

#188 • Braking Systems

Why must stored hydraulic pressure be released before opening lines on a hydro-boost or ABS accumulator system?

Answer

Hydro-boost accumulators, electro-hydraulic units, and high pressure ABS accumulators can hold fluid at very high pressure even with the engine off and battery disconnected. Opening a line on a pressurized system can cause a dangerous fluid spray. Depressurizing is commonly done by pumping the brake pedal firmly 25 to 40 times with the ignition off, or by running a scan tool depressurization routine.

#189 • Braking Systems

What safety rule applies to gas-charged brake accumulators?

Answer

Gas-charged accumulators contain a nitrogen precharge and must never be punctured, heated, or discarded while intact. They must be disposed of following the manufacturer's specified procedure and OSHA safe work practices. Every accumulator should be treated as charged until it is proven otherwise.

#190 • Braking Systems

How can a vacuum or hydraulic assist problem be told apart from an air or hydraulic leak problem by pedal feel?

Answer

Assist problems, such as a weak booster or failing hydro-boost pump, change how much effort the pedal takes without changing pedal height. Air in the lines or a hydraulic leak instead changes pedal height and feel, often producing a low or spongy pedal. This distinction helps a technician quickly narrow down where to start diagnosing.

#191 · Braking Systems

Why can one corroded wheel speed sensor tone ring cause several different warning lamps at once?

Answer

Wheel speed sensors feed data to the ABS module, the traction control system, the stability control system, and EBD, all of which rely on the same signals. A single corroded or damaged tone ring sending an erratic signal can be read by the module as a wheel approaching lockup. This one fault can trigger a false ABS activation, an EBD complaint, and a traction control warning light all together.

#192 · Braking Systems

On late model vehicles with EBD, why should rear brake lockup be diagnosed as an ABS system fault rather than a mechanical proportioning valve problem?

Answer

Most modern vehicles have replaced the mechanical proportioning valve entirely with electronic brake force distribution running through the ABS hydraulic control unit. There is no separate proportioning valve left to inspect or replace on these vehicles. A rear lockup complaint on a late model vehicle should instead point the technician toward checking for ABS system faults.

#193 • Electrical Systems

What is voltage?

Answer

Voltage is electrical pressure, measured in volts. It is the force that pushes electrons through a circuit, similar to how water pressure pushes water through a pipe. Higher voltage means a stronger push on the electrons.

#194 • Electrical Systems

What is current?

Answer

Current is the flow of electrons through a circuit, measured in amperes (amps). More current means more electrons are moving past a point each second. Loads like motors and lamps need a certain current to work correctly.

#195 · Electrical Systems

What is resistance?

Answer

Resistance is opposition to current flow, measured in ohms. Corroded terminals, thin wires, and loads such as bulbs or motors all add resistance. More resistance in a circuit means less current can flow for a given voltage.

#196 · Electrical Systems

State Ohm's law.

Answer

Ohm's law says voltage equals current multiplied by resistance, written $V = I \times R$. It can be rearranged to find current ($I = V / R$) or resistance ($R = V / I$). Exam questions often give two values and ask you to solve for the third.

#197 · Electrical Systems

A 12 volt circuit feeds a lamp with 4 ohms of resistance. What is the current?

Answer

Current equals voltage divided by resistance, so 12 divided by 4 equals 3 amps. This is a direct application of $I = V / R$. If resistance rises later, current will drop for the same 12 volt supply.

#198 · Electrical Systems

Why does a corroded connection make a lamp dim?

Answer

Corrosion adds unwanted resistance in the circuit. Added resistance reduces the current that reaches the lamp, so the lamp glows weaker than normal. This same idea explains most electrical complaints in the brake and suspension trade.

#199 · Electrical Systems

How do you calculate electrical power?

Answer

Power equals current multiplied by voltage, written $P = I \times V$, and is measured in watts. For example, a blower motor drawing 15 amps at 14 volts uses 15 times 14, which is 210 watts. Power calculations matter for sizing fuses and judging normal current draw.

#200 · Electrical Systems

What is a series circuit?

Answer

A series circuit has only one path for current, so the same current flows through every component. Resistances add together, and the source voltage divides across the loads based on their resistance. If any one component opens, the whole circuit stops working.

#201 · Electrical Systems

Two resistors, 2 ohms and 4 ohms, are wired in series on a 12 volt supply. Find total current.

Answer

Total resistance in series is the sum, so 2 plus 4 equals 6 ohms. Current is voltage divided by resistance, 12 divided by 6, which is 2 amps. This same 2 amps flows through both resistors since it is a series circuit.

#202 · Electrical Systems

What is a parallel circuit?

Answer

A parallel circuit gives current more than one path to follow. Voltage is the same across every branch, but each branch draws its own current based on its own resistance. Total resistance is always lower than the smallest branch resistance.

#203 · Electrical Systems

A 6 ohm branch and a 3 ohm branch are wired in parallel across 12 volts. Find the current in each branch.

Answer

Since voltage is the same across both branches, use $I = V / R$ for each one. The 6 ohm branch draws 12 divided by 6, which is 2 amps, and the 3 ohm branch draws 12 divided by 3, which is 4 amps. Total current from the source is 2 plus 4, which is 6 amps.

#204 · Electrical Systems

Why is vehicle lighting usually wired in parallel?

Answer

In a parallel circuit each branch has its own current path and the same source voltage. If one bulb fails open, only that branch loses power and the other bulbs keep working. This is why one burned out bulb does not kill the rest of the lighting circuit.

#205 · Electrical Systems

What is a series-parallel circuit?

Answer

A series-parallel circuit combines both wiring styles in one network. Part of the circuit shares one current path in series, while another part splits into parallel branches. A dash dimmer feeding several parallel bulbs through one series control is a common example.

#206 · Electrical Systems

What is an open circuit fault?

Answer

An open circuit is a break in the current path, caused by things like a broken wire, blown fuse, failed switch, or unplugged connector. Current stops completely and the load goes dead. Full source voltage can usually be measured up to the break, and zero after it.

#207 · Electrical Systems

What is a short to ground?

Answer

A short to ground is an unintended path from the power side of a circuit straight to chassis ground before the load. Because the load's resistance is bypassed, current rises sharply and blows the fuse. A circuit that repeatedly blows fuses is the classic sign of a short to ground.

#208 · Electrical Systems

What is a short to power (short to voltage)?

Answer

A short to power is a copper-to-copper contact between two separate circuits, so one circuit back-feeds another. Symptoms are strange, such as brake lights turning on with the parking lights. Fuses often survive because the load in the other circuit still limits current.

#209 · Electrical Systems

What is a high resistance fault and why is it hard to find?

Answer

High resistance is unwanted resistance added by corrosion, loose terminals, frayed strands, or poor grounds. The circuit still works but weakly, showing symptoms like dim lamps, slow motors, or erratic sensor readings. It does not blow fuses, so it hides from casual inspection and needs voltage drop testing to find.

#210 · Electrical Systems

What does voltage drop testing measure?

Answer

Voltage drop testing measures the voltage lost across a connection, wire, or switch while the circuit operates under normal load. You connect the voltmeter across the suspected section and operate the circuit. Whatever voltage the meter reads is being wasted there instead of powering the load.

#211 · Electrical Systems

Why is voltage drop testing better than a simple ohmmeter check?

Answer

An ohmmeter pushes only a tiny current through the circuit, so a corroded connection can read a low resistance yet still fail under real load. Voltage drop testing loads the circuit with its actual operating current, which reveals marginal connections. It also tests the circuit without taking anything apart.

#212 · Electrical Systems

What are the accepted voltage drop limits for a connection and a switch?

Answer

A single connection or junction should drop no more than about 0.2 volt, and a wire or cable run should also stay near 0.2 volt. A switch is allowed a bit more, about 0.3 volt. Readings above these limits point to unwanted resistance.

#213 · Electrical Systems

What voltage drop limit applies to the starter circuit during cranking?

Answer

Both the positive cable path and the ground path on the starter circuit should each stay at or below about 0.5 volt during cranking. A higher reading on either side points to a bad connection or corroded cable in that path. This is a high current circuit, so even small resistance causes a real voltage loss.

#214 · Electrical Systems

What does a voltage drop reading near source voltage across a section mean?

Answer

A voltage drop reading close to full source voltage across a section means that section is open. Almost all the source voltage is being lost right there instead of reaching the load. This is a classic exam logic trap worth memorizing.

#215 · Electrical Systems

How do you connect a DMM to measure voltage?

Answer

Connect the meter leads in parallel with, meaning across, the component or section being tested. Select DC volts for vehicle circuits and read the display. Auto-ranging meters pick the scale automatically, but always check the units shown, since 200 mV is only 0.2 volt.

#216 · Electrical Systems

How do you connect a DMM to measure current, and why is this risky on a starter circuit?

Answer

To measure current the meter must be connected in series so the circuit's current flows through it, which means opening the circuit. Most DMMs are fused at only 10 amps on the direct amps input. Putting the meter in series with a starter circuit, which can draw over 100 amps, will blow the meter fuse instantly or damage the meter.

#217 · Electrical Systems

What is an inductive amp clamp used for?

Answer

An inductive amp clamp measures current by sensing the magnetic field around a wire, without breaking the circuit. This avoids the risk of putting a fused meter directly in series with a high current circuit. It is the safer choice for measuring starter or charging current.

#218 · Electrical Systems

What must you do before measuring resistance with a DMM?

Answer

The circuit must be electrically dead and the component isolated before checking resistance. Never connect an ohmmeter to a live circuit, because outside voltage will give a false reading and can damage the meter. Remember that an ohmmeter's low test current also cannot find load-sensitive high resistance faults.

#219 · Electrical Systems

What does the min/max function on a DMM do, and when is it useful?

Answer

The min/max function automatically records the highest and lowest readings the meter sees. It is ideal for catching intermittent faults, such as setting it on a sensor's supply voltage and wiggling the harness. The meter memorizes any dropout even if it lasts only a moment.

#220 · Electrical Systems

Why do good automotive DMMs have high input impedance?

Answer

High input impedance, typically 10 megohms or more, means the meter draws almost no current from the circuit being tested. This prevents the meter from skewing delicate computer circuits during a measurement. A low impedance meter could load down a sensitive signal and give a false reading.

#221 • Electrical Systems

Why is a conventional test light dangerous on computer circuits?

Answer

A conventional test light is a bulb with a probe and ground clip, and it can pull enough current to damage a computer driver circuit. It should never be used to probe air bag (SRS) circuits or low-current sensor and module circuits. A high-impedance logic probe is the safer tool for that job.

#222 • Electrical Systems

What is a logic probe used for?

Answer

A logic probe, or high-impedance LED test light, draws almost no current and displays high, low, or pulsing signals. It is the safe choice for checking signals around modules. Its pulse indication can confirm that an injector driver or ABS solenoid driver is switching correctly.

#223 · Electrical Systems

When is a lab scope the correct tool instead of a meter?

Answer

A lab scope is needed when the actual shape of a signal matters, not just its presence. A wheel speed sensor waveform is a good example, since a cracked tone ring shows up as a distorted shape rather than a simple voltage change. A meter cannot show this kind of detail.

#224 · Electrical Systems

What do the symbols on a wiring diagram represent?

Answer

A wiring diagram uses standard symbols such as a sawtooth or rectangle for a resistor, a circle with a filament for a lamp, parallel long and short lines for a battery, and angled contact drawings for switches. Ground is usually shown as a set of descending lines or a triangle. Wires are also labelled with circuit numbers, gauge, and colour codes.

#225 · Electrical Systems

What does a connector and pin number reference, like C204 pin 14, tell you?

Answer

Connectors are numbered, such as C204, and each cavity or pin inside a connector has its own pin number. This lets a wiring diagram direct a technician to an exact point in a harness, such as pin 14 of connector C204. It removes guesswork when probing a specific wire.

#226 · Electrical Systems

What order should you follow when tracing a circuit on a wiring diagram?

Answer

Start at the power source and follow through the protection device, then the switch or relay, then the load, and finally the ground point. This top to bottom power flow matches how most diagrams are drawn. Planning test points that split the circuit in half helps each measurement eliminate as much wiring as possible.

#227 · Electrical Systems

What rule applies when replacing a blown fuse?

Answer

Always replace a fuse with one of the same rating, never a higher rating. Blade fuses are colour coded so the correct rating is easy to identify. A fuse that blows again means there is a circuit problem to find, not a fuse problem.

#228 · Electrical Systems

What is a fusible link and what does it protect?

Answer

A fusible link is a short length of wire, typically four gauge sizes smaller than the wire it protects, with special insulation that blisters when the link burns. Fusible links protect high current feeds such as the charging and starting circuits. A melted, swollen, or crusty section of harness near the battery is the classic sign of a blown link.

#229 · Electrical Systems

What is a cycling (self-resetting) circuit breaker used for?

Answer

A cycling circuit breaker opens on overload but resets itself once it cools, allowing it to protect circuits where a temporary overload is normal. Headlamps, power windows, and wipers commonly use this type. A headlamp that flashes on and off rhythmically suggests a cycling breaker responding to a partial short.

#230 · Electrical Systems

On a standard ISO relay, what do terminals 30 and 87 do?

Answer

Terminal 30 is the main power feed into the relay. Terminal 87 is the output that sends power on to the load once the relay is energized. Energizing the coil connects 30 to 87 internally.

#231 · Electrical Systems

On a standard ISO relay, what do terminals 85 and 86 do?

Answer

Terminals 85 and 86 are the two ends of the control coil inside the relay. Applying power and ground to these terminals energizes the coil, which pulls the main contacts closed. This lets a small control current switch a much larger load current at terminals 30 and 87.

#232 · Electrical Systems

What is the typical coil resistance range for a relay, and why do many relays include a diode?

Answer

Relay coil resistance is typically about 50 to 120 ohms. Many relays include a clamping diode or resistor across the coil to absorb the voltage spike created when the coil's magnetic field collapses. Installing a diode relay backwards will short the control circuit, so replacement relays must match exactly.

#233 · Electrical Systems

What is the difference between a rheostat and a potentiometer?

Answer

A rheostat has two terminals and varies current directly, as used in a dash dimmer. A potentiometer has three terminals and divides a reference voltage into a smooth output signal, as used in many pedal position sensors. A potentiometer's output should sweep smoothly with no dropouts.

#234 · Electrical Systems

How many cells does a lead-acid battery have, and what is its fully charged voltage?

Answer

A lead-acid battery has six cells in series, each producing about 2.1 volts. This gives a fully charged open circuit voltage of about 12.6 volts or slightly higher. Each cell holds lead plates in an electrolyte of sulphuric acid and water.

#235 · Electrical Systems

What does cold cranking amps (CCA) measure?

Answer

Cold cranking amps is the current a battery can deliver for 30 seconds at minus 18 degrees Celsius while staying above 7.2 volts. It rates how well a battery can crank an engine in cold weather. A higher CCA rating generally means stronger cold weather starting performance.

#236 · Electrical Systems

What does reserve capacity (RC) measure?

Answer

Reserve capacity is the number of minutes a battery can supply 25 amps before its voltage drops to 10.5 volts. It estimates how long a vehicle can keep running on the battery alone if the charging system fails. It is a different measurement than CCA, which focuses on cold cranking strength instead.

#237 · Electrical Systems

What open circuit voltage corresponds to a 50 percent charged battery?

Answer

An open circuit voltage of about 12.2 volts corresponds to roughly a 50 percent state of charge after the battery has rested. About 12.6 volts is fully charged, 12.4 volts is about 75 percent, and 12.0 volts is about 25 percent. Surface charge must be removed first or the reading will be misleadingly high.

#238 · Electrical Systems

How do you remove surface charge before checking open circuit voltage?

Answer

Turn on the headlamps for about a minute, or crank the engine briefly, then let the battery sit before reading its voltage. Recent charging or driving leaves a surface charge that reads higher than the battery's true state. Skipping this step gives a misleadingly high voltage reading.

#239 · Electrical Systems

How is a battery load test performed and what result is expected?

Answer

A load test applies half the battery's CCA rating for 15 seconds. A good battery at room temperature should stay above about 9.6 volts during that test. The battery must be at or near full charge before the test, or the result is meaningless.

#240 · Electrical Systems

What is a conductance tester and why is it common in shops?

Answer

A conductance tester sends a small AC signal through the battery and estimates its capacity electronically. It is safe, fast, and works even on a partly discharged battery. This makes it the standard first test in most shops today, ahead of a traditional load test.

#241 · Electrical Systems

Why should you never charge or jump start a frozen battery?

Answer

Ice inside a frozen battery can crack the case, and the battery can rupture if charged or jump started while frozen. The safe procedure is to let it thaw completely and inspect it first. Charging should also always happen in a ventilated area since it produces explosive hydrogen gas.

#242 · Electrical Systems

What makes an AGM battery different, and what charging precaution does it need?

Answer

An absorbed glass mat (AGM) battery holds its electrolyte in fibreglass separators instead of free liquid, which resists vibration and tolerates deep cycling better. It needs a charger with an AGM setting because sustained charging much above about 14.8 volts damages it. Treating it like a standard flooded battery can shorten its life.

#243 · Electrical Systems

What does a memory saver do when disconnecting a battery, and what precaution still applies?

Answer

A memory saver keeps modules powered and preserves settings while the main battery is disconnected. Even so, the vehicle should be treated as fully powered while the saver is connected, since circuits remain live. An accidental short through the memory saver path can damage it or the vehicle's modules.

#244 · Electrical Systems

What current draw is typical during starter cranking, and what does higher than normal draw with slow cranking suggest?

Answer

A typical four or six cylinder engine draws roughly 125 to 200 amps once the starter is spinning. Higher than normal current draw combined with slow cranking suggests a dragging starter or excessive engine resistance. This is different from a resistance problem in the wiring, which shows the opposite pattern.

#245 · Electrical Systems

What does lower than normal starter current draw with slow cranking suggest?

Answer

Lower than normal current draw with slow cranking points to high resistance somewhere in the cables or connections. The starter is being starved of current rather than dragging mechanically. Voltage drop testing on the positive and ground cables will locate the exact resistance.

#246 · Electrical Systems

What system voltage indicates a healthy charging system at 2,000 rpm under load?

Answer

A healthy alternator holds system voltage in roughly the 13.5 to 14.8 volt range at about 2,000 rpm with loads applied. Readings persistently below about 13.2 volts indicate undercharging. Readings above about 15 volts indicate overcharging, which can cook batteries and modules.

#247 · Electrical Systems

What does excess AC ripple from an alternator indicate?

Answer

Alternators produce AC internally and rectify it to DC through diodes, so a failed diode lets AC ripple leak into the system. Excess ripple, checked on the meter's AC volts setting or viewed on a scope, causes flickering lights and strange module behaviour. This condemns the alternator even when the DC voltage reading looks normal.

#248 · Electrical Systems

How does a passive two-wire wheel speed sensor work?

Answer

A passive wheel speed sensor is a small AC generator made of a coil and magnet mounted beside a toothed tone ring. As the teeth pass, the sensor produces an AC voltage whose amplitude and frequency rise with wheel speed. It reads poorly at very low speeds, unlike an active sensor.

#249 · Electrical Systems

How does an active three-wire wheel speed sensor differ from a passive one?

Answer

An active sensor uses a Hall effect or magneto-resistive element with its own power supply, producing a clean digital square wave. It stays accurate down to nearly zero speed, unlike a passive sensor. Ordinary resistance checks do not apply to an active sensor and can mislead a technician.

#250 · Electrical Systems

Why must a steering angle sensor usually be recalibrated after an alignment?

Answer

The steering angle sensor reports wheel position, rate, and direction to the stability control system. Most vehicles require recalibration or re-zeroing after an alignment or any steering component work. Skipping this reset can trigger stability control faults or false activations.

#251 • Electrical Systems

What is the expected CAN bus resistance reading at the diagnostic connector, and what does a reading near 120 ohms mean?

Answer

With the ignition off and battery disconnected, the two 120 ohm terminating resistors in parallel should read about 60 ohms across the CAN high and CAN low pair. A reading near 120 ohms means one terminating resistor or its section of bus is missing. Readings far from 60 ohms point to shorted or open bus wiring.

#252 • Electrical Systems

Why must you disconnect the battery and wait before working near SRS wiring, and what electrical mistake can deploy an air bag?

Answer

The supplemental restraint system (SRS) stores electrical energy to fire air bags and pretensioners even after the battery is disconnected. Waiting the manufacturer's specified time, commonly one to three minutes, lets the reserve capacitor discharge safely. Never probe SRS circuits with a test light or ohmmeter, since the small current involved can be enough to deploy a device.

#253 · Safety Work and Business Practices

What is the correct order of the hierarchy of hazard controls?

Answer

The order is elimination first, then substitution, then engineering controls, then administrative controls, and PPE last. PPE is the last line of defence, not the first choice. An exam question about controlling solvent vapour is usually answered by substitution or ventilation, not by handing out respirators.

#254 · Safety Work and Business Practices

Why should a technician never blow brake dust off with compressed air?

Answer

Old friction materials may contain asbestos fibres, which cause serious lung disease years after exposure. Compressed air, dry brushing, and dry sanding all lift the finest fibres into the breathing zone. Since a technician cannot tell by looking whether a part is asbestos free, all brake dust must be treated as if it contains asbestos.

#255 · Safety Work and Business Practices

Name the two approved methods for cleaning brake dust from an assembly.

Answer

The first is a HEPA-filtered vacuum enclosure system that captures dust through a filter fine enough to trap asbestos fibres. The second is the wet wash or low-pressure wetting method, where water with a wetting agent or aerosol brake cleaner turns the dust into a slurry. An ordinary shop vacuum is not acceptable because it passes fine fibres back into the air.

#256 · Safety Work and Business Practices

What Ontario regulation classifies wet brake servicing as a Type 1 asbestos operation?

Answer

Ontario Regulation 278/05 classifies brake servicing that uses wet methods or enclosure as a Type 1 asbestos operation with defined precautions. This means the shop must follow specific handling and disposal steps. Wet rags, filters, and slurry are sealed in labelled containers rather than thrown in general garbage.

#257 · Safety Work and Business Practices

What replaced MSDS and old circular symbols under WHMIS 2015?

Answer

WHMIS 2015 replaced Material Safety Data Sheets with Safety Data Sheets and replaced the old circular symbols with red diamond-bordered GHS pictograms. This change aligned Canada with the Globally Harmonized System used internationally. WHMIS still rests on labels, safety data sheets, and worker education.

#258 · Safety Work and Business Practices

What must a workplace label show when solvent is poured from a supplier container into a spray bottle?

Answer

The workplace label needs at least the product name, safe handling information, and a reference to the SDS. An unlabelled spray bottle of mystery solvent is a WHMIS violation and a common exam trap. The employer is responsible for providing labels and SDS access, and workers must follow the training given.

#259 · Safety Work and Business Practices

What should a technician do before raising a vehicle to full height on a hoist?

Answer

Raise the vehicle about 150 to 300 mm first, then shake it to confirm stable, even contact on all arms or pads before going higher. Lift points must match the manufacturer's designated frame, pinch weld, or subframe locations. Skipping this check risks a fall if the pads are not seated correctly.

#260 · Safety Work and Business Practices

Can a vehicle be worked on while resting only on hydraulic jack pressure?

Answer

No, a hydraulic jack is a lifting device only, never a support device. Once raised, the vehicle must be lowered onto safety stands rated for the load before anyone works underneath. Working under a vehicle held only by a jack is one of the classic fatal shortcuts in this trade.

#261 · Safety Work and Business Practices

What are the three core worker rights under Ontario's OHSA?

Answer

Workers have the right to know about workplace hazards, the right to participate in health and safety through committees or representatives, and the right to refuse unsafe work. These rights sit inside the internal responsibility system, where everyone shares safety duties according to their authority. The Act is enforced by the Ministry of Labour, Immigration, Training and Skills Development.

#262 · Safety Work and Business Practices

What size shop requires a joint health and safety committee versus a representative under OHSA?

Answer

Workplaces with 20 or more workers need a joint health and safety committee. Smaller shops with 6 to 19 workers need a health and safety representative instead. This structure supports the worker's right to participate in health and safety matters.

#263 · Safety Work and Business Practices

Why should welding never be done near brake cleaner residue?

Answer

Many aerosol brake cleaners are highly flammable, and even non-flammable chlorinated formulas can decompose under welding heat into phosgene, a highly toxic gas. A part cleaned with chlorinated solvent must be fully dry and ventilated before any welding nearby. This hazard is separate from the normal fire triangle risk of flammable liquids.

#264 · Safety Work and Business Practices

What does the PASS method describe and what class of extinguisher is common in a brake shop?

Answer

PASS stands for pull, aim, squeeze, sweep, describing how to operate a fire extinguisher aimed at the base of the fire. The common shop unit is the multi-purpose ABC dry chemical extinguisher, which covers ordinary combustibles, flammable liquids, and energized electrical equipment. Water is never used on a flammable liquid fire since it spreads the burning fuel.

#265 · Safety Work and Business Practices

How should used brake fluid be handled as waste?

Answer

Used brake fluid is a hazardous waste collected in a dedicated, labelled container and sent to a licensed recycler or hazardous waste hauler. It should not be mixed with used engine oil unless the recycler specifically accepts that blend. Dumping it down a drain or into general garbage violates Ontario environmental law.

#266 · Safety Work and Business Practices

What steps disable the SRS before working on a steering column?

Answer

Turn the ignition off, disconnect the negative battery cable, and wait the manufacturer's specified time, commonly one to three minutes and sometimes up to ten, for the backup capacitor to discharge. This stored energy must be neutralized before touching the clock spring, steering angle sensor, or column. A removed airbag module is carried trim side away from the body and stored trim side up.

#267 · Safety Work and Business Practices

Under Ontario's Consumer Protection Act, by how much can a final repair invoice exceed the written estimate?

Answer

The final invoice cannot exceed the estimate by more than 10 percent without new customer authorization. A written estimate must be given before work starts unless the customer declines one and authorizes a maximum amount. Repairs also carry a minimum warranty of 90 days or 5,000 kilometres, whichever comes first.

#268 · Steering Systems

What are the three jobs the steering system must perform?

Answer

It must convert steering wheel rotation into side-to-side movement of the front wheels, give the driver accurate road feel, and multiply effort so the wheels can be turned at parking speeds. It must also hold the wheels on course without constant correction. Almost every steering complaint traces back to one of these three jobs failing.

#269 · Steering Systems

What are the two basic steering gear layouts used on cars and light trucks?

Answer

The older layout uses a conventional steering gear, usually recirculating ball, driving a parallelogram linkage of arms and rods. The newer layout is rack and pinion, where the gear and the cross-vehicle link are combined into one compact assembly. Most modern front-wheel-drive cars use rack and pinion.

#270 · Steering Systems

Which linkage components belong only to a parallelogram steering system?

Answer

Pitman arms, idler arms, and center links belong only to the parallelogram system. Inner tie rod sockets and bellows boots belong only to rack and pinion instead. Outer tie rod ends appear in both systems.

#271 · Steering Systems

Name the three forms steering assist can take.

Answer

Assist can be hydraulic, using an engine-driven pump. It can be electro-hydraulic, where an electric motor drives the pump. It can also be fully electric, using a motor acting on the column, pinion, or rack.

#272 · Steering Systems

What does the steering column do?

Answer

The steering column carries driver input from the steering wheel down to the steering gear. It also houses the ignition lock, tilt and telescope mechanisms, and the clock spring. Most modern columns are also designed to collapse safely in a crash.

#273 · Steering Systems

How do collapsible steering columns protect the driver in a frontal collision?

Answer

Shear capsules, mesh sections, or telescoping tubes crush in a controlled way during impact. This keeps the column from spearing into the driver. A column that has been in a collision must always be inspected for partial collapse.

#274 · Steering Systems

What tool must be used to remove a steering wheel from its shaft, and why?

Answer

A proper puller must always be used. You should never hammer on the end of a steering shaft or apply heavy axial force during removal. Hammering or forcing the shaft can damage the collapsible column or internal components.

#275 · Steering Systems

What connects the steering column to the steering gear when they do not line up?

Answer

An intermediate shaft with one or two universal joints, or a flexible coupling, connects the column to the gear. This bridges the gap between the two components. It allows for slight misalignment in the vehicle's packaging.

#276 · Steering Systems

Why are dual universal joints on an intermediate shaft phased to each other?

Answer

Phasing the two joints cancels out each joint's speed variation as the shaft rotates. A shaft installed out of phase causes a cyclic bind or surging effort as the driver turns the wheel. Correct phasing keeps steering effort smooth through a full turn.

#277 · Steering Systems

What symptom points to a worn intermediate shaft joint?

Answer

A worn joint shows up as a small amount of free play at the steering wheel that does not appear at the linkage. It can also feel like a clunk in the column over bumps. This distinguishes it from linkage or gear wear, which shows up differently.

#278 · Steering Systems

What commonly causes a wobbly steering wheel that has nothing to do with the gear or linkage?

Answer

Loose tilt pivot pins in the steering column are a classic cause of this complaint. The wobble originates in the column mechanism itself rather than downstream in the gear or linkage. Checking the tilt mechanism should be part of any wobble diagnosis.

#279 · Steering Systems

What is a clock spring, and what does it do?

Answer

A clock spring, also called a spiral cable, is a flat ribbon conductor wound inside a housing at the top of the column. It keeps the air bag, horn, and steering wheel switches electrically connected while the wheel turns lock to lock. It has a limited number of turns in each direction.

#280 · Steering Systems

Before working on the steering column or air bag, what safety step must be taken first?

Answer

Disconnect the battery negative cable and wait the manufacturer's specified time. This is commonly at least one minute and sometimes as long as ten minutes. The wait allows the SRS reserve capacitor to discharge safely.

#281 · Steering Systems

Before disconnecting the steering linkage or intermediate shaft, what two steps protect the clock spring?

Answer

Set the front wheels straight ahead first, then lock the steering wheel with the ignition lock or a wheel holder so it cannot spin freely. This prevents the disconnected wheel from being spun by accident. Never let a disconnected shaft rotate with the linkage detached.

#282 · Steering Systems

What happens if a disconnected steering wheel is spun freely?

Answer

Spinning the wheel winds the clock spring past its travel limit and snaps the ribbon conductor inside it. This sets an air bag warning lamp and usually kills the horn and cruise control switches. A broken clock spring must then be replaced and properly centered.

#283 · Steering Systems

What must be done to a replacement clock spring before the steering wheel is installed?

Answer

The replacement clock spring must be centered according to the service information. This ensures the ribbon has equal travel in both directions once the wheel is reinstalled. Skipping this step risks snapping the new clock spring the same way as the old one.

#284 · Steering Systems

What locks the steering shaft when the ignition key is removed?

Answer

The ignition lock, housed in the steering column, locks the shaft when the key is removed. This is a theft deterrent built into the column assembly. It works alongside the tilt and telescope mechanisms that also live in the column.

#285 · Steering Systems

What is a recirculating ball steering gear?

Answer

It is a conventional steering gear where the steering shaft turns a worm gear, and a ball nut rides on the worm. Steel balls circulate between the nut and worm through guide tubes instead of the threads touching directly. This design gives very low friction compared to a plain worm and sector gear.

#286 · Steering Systems

How does motion get from the ball nut to the pitman arm in a recirculating ball gear?

Answer

Teeth on the side of the ball nut mesh with a sector gear on the pitman shaft. The pitman shaft rocks back and forth as the ball nut moves. This rocking motion moves the pitman arm and the connected linkage.

#287 · Steering Systems

What is the older gear type that works like a recirculating ball gear but with more friction?

Answer

A plain worm and sector gear, with no circulating balls, works on the same principle. Without the ball bearings reducing friction, it requires more effort to turn. This design is mostly found on older equipment.

#288 · Steering Systems

What are the two adjustments on a recirculating ball steering gear?

Answer

Worm bearing preload sets the drag on the bearings supporting the worm shaft. Sector lash, also called overcenter adjustment, sets the clearance between the ball nut teeth and the sector teeth. Both adjustments are needed to keep the gear tight without binding.

#289 · Steering Systems

How is worm bearing preload measured on a recirculating ball gear?

Answer

It is measured with an inch-pound torque wrench while rotating the input shaft. This reads the drag created by the worm bearings. It must be set before the sector lash adjustment is made.

#290 · Steering Systems

Why must sector lash be adjusted with the gear exactly on center?

Answer

The gear is machined with a high point at its center, which is where it is tightest. Adjusting lash anywhere else would not reflect the true tightest point of travel. Setting lash off-center can leave excess play once the wheels return to straight ahead.

#291 · Steering Systems

What is the correct adjustment order for a recirculating ball steering gear, and why does it matter?

Answer

Worm bearing preload must always be set first, then sector lash is set second. If lash is set first and preload is changed afterward, the lash setting gets disturbed. This order is a common exam trap on the 310E exam.

#292 · Steering Systems

Why are recirculating ball gear adjustments made with the linkage disconnected?

Answer

Disconnecting the linkage removes its drag from the readings. If the linkage stayed connected, its resistance could mask the true preload or lash values. This keeps the adjustment accurate to the gear alone.

#293 · Steering Systems

What symptom points to excessive sector lash in a steering gear?

Answer

Excessive lash shows up as steering wheel free play. This free play is worst when the wheel is in the straight-ahead position. It happens because straight ahead is the gear's tightest, highest point.

#294 · Steering Systems

Why is a parallelogram linkage called by that name?

Answer

It is named because the center link and the line between the two arm pivots stay roughly parallel as the linkage swings through its travel. This geometric relationship is what defines the linkage type. It applies only to vehicles using a conventional steering gear.

#295 · Steering Systems

What is the function of the pitman arm in a parallelogram linkage?

Answer

The pitman arm bolts to the splined pitman shaft of the steering gear. It provides the input motion for the entire linkage. As the gear rocks the pitman shaft, the arm swings and moves the center link.

#296 · Steering Systems

What is the function of the idler arm in a parallelogram linkage?

Answer

The idler arm bolts to the frame on the opposite side of the vehicle from the pitman arm. It supports the other end of the center link at the same height. This keeps the center link level as it moves side to side.

#297 · Steering Systems

What is the center link, and what other names does it go by?

Answer

The center link spans the vehicle between the pitman arm and the idler arm. It is also called a relay rod or a drag link. It is the main cross-vehicle member of a parallelogram linkage.

#298 · Steering Systems

What connects the center link to the steering knuckle on each side?

Answer

An inner tie rod end, an adjuster sleeve, and an outer tie rod end connect the center link to the steering arm at each knuckle. The adjuster sleeve allows the tie rod length to be changed for toe adjustment. Its clamp must be positioned so it never contacts anything through the full range of travel.

#299 · Steering Systems

What is the most common parallelogram linkage failure?

Answer

A worn idler arm is the most common parallelogram failure. It causes wander and toe change while driving. Because the idler arm supports the center link, wear there affects the whole linkage's geometry.

#300 · Steering Systems

How is idler arm deflection checked?

Answer

A specified load, commonly 110 N or 25 lb, is applied up and then down at the center link end of the idler arm. The total vertical movement is measured. This is a load-and-measure check, not a shake test, because an unloaded idler arm can feel fine and still be out of spec.

#301 · Steering Systems

What is a typical allowable total deflection for an idler arm?

Answer

A typical allowable total deflection is about 3 mm, or 1/8 inch. Anything beyond the manufacturer's specification means the idler arm must be replaced. The exact figure should always be confirmed in the service information.

#302 · Steering Systems

How is a pitman arm socket checked for wear?

Answer

The pitman arm socket is checked for any perceptible looseness while under load. Most manufacturers allow essentially no play in this joint. Any detectable movement usually means the pitman arm needs replacement.

#303 · Steering Systems

What must be done after replacing any joint in a parallelogram linkage?

Answer

Toe must be reset on the alignment rack whenever any linkage joint is replaced. Replacing a joint changes the effective length or geometry of the linkage. Skipping this step leaves the vehicle out of alignment even if the new part is installed correctly.

#304 · Steering Systems

How are outer tie rod ends typically checked for wear?

Answer

The vehicle is loaded or supported in its normal ride position, then the tire is rocked at the 3 and 9 o'clock positions. A technician can also push and pull the joint directly while watching for movement between the stud and socket. Any perceptible looseness in a follower-type tie rod end usually means replacement.

#305 · Steering Systems

Does every tie rod end design use the same looseness standard?

Answer

No, some designs specify a small allowable amount of movement rather than zero. A technician should always check the service information for the exact specification. Assuming any looseness means failure can lead to unnecessary replacement.

#306 · Steering Systems

Why does a torn boot condemn a tie rod end?

Answer

A torn boot lets grease escape and allows contamination such as water and grit to enter the joint. This quickly wears out the internal socket. Once the boot is torn, the joint is considered failed even if it currently feels tight.

#307 · Steering Systems

What two methods keep a replacement outer tie rod end close to its original length?

Answer

One method is marking the old position and counting the exact number of turns needed to remove it, then threading the new end the same number of turns. The other method, used on parallelogram systems, measures the distance between grease fittings or fixed reference points on the old assembly. Both approximate the original length so the vehicle can be safely driven to the alignment rack.

#308 · Steering Systems

Are these tie rod length matching methods a substitute for a wheel alignment?

Answer

No, they are not a substitute for alignment. They only return the tie rod to approximately its original length so toe is close enough for a safe drive. Toe must still be measured and reset on the alignment rack afterward.

#309 · Steering Systems

What tool should be used to separate a tie rod end from the steering knuckle?

Answer

A proper puller or tie rod separator should be used on the tapered stud. Hammering the stud sideways damages the knuckle taper. A pickle fork tears the boot, so it is only acceptable when the joint is being discarded anyway.

#310 · Steering Systems

Why must tie rod length changes always be followed by an alignment check?

Answer

Tie rod length directly sets toe. Any time a tie rod end, adjuster sleeve, or inner socket is loosened or replaced, the effective tie rod length has changed. Toe must then be measured and reset on the alignment rack.

#311 · Steering Systems

What causes an off-center steering wheel right after tie rod work, even when total toe is correct?

Answer

This happens when the tie rods were adjusted unequally on each side. Total toe can be correct while the split between the left and right sides is wrong. The fix is adjusting both tie rods in equal and opposite amounts until the wheel centers with toe still in spec.

#312 · Steering Systems

How does a rack and pinion steering gear convert rotation into wheel movement?

Answer

A pinion gear on the end of the steering shaft meshes directly with a toothed rack bar. Turning the pinion slides the rack left or right inside its housing. The rack ends then push and pull the tie rods to turn the wheels.

#313 · Steering Systems

What are the main advantages of rack and pinion steering over a parallelogram system?

Answer

Rack and pinion has fewer parts and wear points, lower weight, and less friction. It gives more direct road feel and fits in a compact package suited to front-wheel-drive engine bays. The main trade-off is that road shock feeds back to the driver more directly.

#314 · Steering Systems

How is a rack and pinion gear typically serviced when its internal parts wear out?

Answer

The whole unit is usually replaced rather than overhauled. This differs from a recirculating ball gear, which can often be adjusted or rebuilt. Internal rack and pinion wear is generally not repaired piece by piece.

#315 · Steering Systems

What is an inner tie rod socket on a rack and pinion gear, and what protects it?

Answer

It is a ball socket that threads onto the rack bar and lets the tie rod swing with suspension travel. A bellows boot, clamped between the housing and the tie rod, protects the socket from water and grit. A torn boot lets contamination destroy the inner socket quickly.

#316 · Steering Systems

What must be inspected before deciding on repair when a rack and pinion boot is found torn?

Answer

The inner socket must be inspected before any repair decision is made. A torn boot allows water and grit to enter and damage the socket. If the socket is already worn, boot replacement alone will not fix the complaint.

#317 · Steering Systems

How is inner tie rod socket play checked on a rack and pinion gear?

Answer

The boot is squeezed to grip the socket while an assistant rocks the wheel. A technician watches or feels for lost motion between the rack and the tie rod. This checks the socket without needing to remove the boot.

#318 · Steering Systems

What is the outer tie rod end on a rack and pinion system, and how is it secured?

Answer

It is a conventional tapered-stud ball joint. It threads onto the inner tie rod and is secured with a jam nut. This design is essentially the same outer joint used on parallelogram systems.

#319 · Steering Systems

What problems can loose or deteriorated rack mount bushings cause?

Answer

Loose mounts let the whole rack housing shift under steering load. This can feel like a clunk on turn-in, momentary free play, or steering response that differs from side to side. Mount bushings and mounting bolt torque should always be checked before condemning the gear itself.

#320 · Steering Systems

What is the rack guide adjustment, and what happens if it is set wrong?

Answer

The rack guide, also called the rack yoke or rack bearing, is a spring-loaded yoke under an adjuster plug that presses the rack into the pinion teeth to control lash. It is typically tightened to a small torque, backed off a specified angle such as 45 to 60 degrees, then locked. Too tight causes hard steering and poor returnability, while too loose causes rattle over bumps and free play at center.

#321 · Steering Systems

What type of pump does hydraulic power steering almost always use?

Answer

A vane type pump. A rotor with sliding vanes spins inside an elliptical cam ring, drawing fluid in where the chamber grows and pushing it out where the chamber shrinks. This design is compact and delivers steady flow across a wide range of engine speeds.

#322 · Steering Systems

What does the flow control valve inside a power steering pump do?

Answer

It is a spring loaded spool that bypasses excess fluid back to the intake as engine speed rises. This keeps the flow delivered to the steering gear roughly constant no matter how fast the engine is turning. Without it, assist would change with every change in engine rpm.

#323 · Steering Systems

What is the job of the pressure relief ball inside a power steering pump?

Answer

It limits the maximum system pressure the pump can produce. This matters most when the wheel is held hard against a stop, since that is when the pump is forced to relieve at full pressure. Without it, pressure could climb high enough to damage hoses or seals.

#324 · Steering Systems

Why should you never hold a power steering wheel on full lock for very long during testing?

Answer

Holding full lock forces the pump to relieve at maximum pressure continuously. This overheats the fluid and can damage the pump over time. A common rule is not to hold full lock longer than about five seconds during any test.

#325 · Steering Systems

How is a power steering pressure test set up?

Answer

A gauge and shutoff valve are installed in the pressure line between the pump and the steering gear. With the valve open, the engine idling, and the fluid at operating temperature, you read system pressure while turning the wheel. You then briefly close the valve to read the pump's maximum output pressure.

#326 · Steering Systems

What does the restricted test (closing the shutoff valve) during a power steering pressure test tell you?

Answer

It reads the pump's maximum possible pressure output. If that reading is low, the pump or its internal relief valve is at fault. If maximum pressure is fine there, the problem lies somewhere else in the system, usually inside the gear.

#327 · Steering Systems

In a power steering pressure test, pump pressure is good but effort is high when turning against the gear with the valve open. Where is the leak?

Answer

The leak is inside the steering gear itself, usually at the spool valve or the rack piston seals. The pump is proven good by the restricted test, so the problem must be downstream of the pump. This pump versus gear logic is a common exam question.

#328 · Steering Systems

About what pressure range does a power steering pump typically produce at full restriction?

Answer

Roughly 5,500 to 10,300 kPa (800 to 1,500 psi), depending on the vehicle. The exact specification always comes from the manufacturer's service information. A reading well below spec points to a weak pump or a bad relief valve.

#329 · Steering Systems

Why is fluid condition always checked first in power steering diagnosis?

Answer

Many systems use different fluids, some take dedicated power steering fluid, others use automatic transmission fluid, and some require a manufacturer specific synthetic. Mixing incompatible fluids swells the internal seals and causes leaks. Using the wrong fluid can create a problem that looks like a hardware failure.

#330 · Steering Systems

How do you flush a power steering system after a pump or gear failure leaves debris behind?

Answer

Disconnect the return line and cap the reservoir return port so old fluid does not recirculate. Run fresh fluid through the system while an assistant cycles the steering wheel at idle. Continue until clean fluid comes out, then reconnect everything and bleed the system.

#331 · Steering Systems

What power steering symptom does a glazed or loose drive belt cause?

Answer

An intermittent loss of assist, often noticed most during slow parking maneuvers. The belt slips under the higher load of a hard turn, so the pump cannot keep up. Checking belt condition and tension is a basic early step in hydraulic power steering diagnosis.

#332 · Steering Systems

What are the signs of aerated power steering fluid?

Answer

The pump whines or growls, the fluid in the reservoir looks foamy or milky, and steering assist becomes jerky. The cause is usually a low fluid level or an air leak on the suction side of the pump, such as a loose intake hose clamp or a bad reservoir O-ring. Aeration is the classic power steering noise complaint.

#333 · Steering Systems

What is cavitation in a power steering pump and what causes it?

Answer

Cavitation is vapor bubbles forming and then collapsing because the pump is starved for fluid. It produces a rattling growl similar to aeration noise. The usual cause is a restricted suction hose or a clogged reservoir screen.

#334 · Steering Systems

How do you bleed a power steering system after repairs?

Answer

Raise the front wheels off the ground and cycle the steering from lock to lock following the service procedure. Keep topping up the fluid reservoir until the level holds steady and the foam clears. This removes trapped air introduced during the repair.

#335 · Steering Systems

What does morning sickness mean in power rack and pinion diagnosis?

Answer

It is the trade name for a power rack that steers hard for the first few minutes on a cold morning, then works normally once it warms up. The cause is worn Teflon spool valve rings that have cut grooves into the aluminum valve housing bore, letting cold thick fluid bypass. The fix is a replacement rack or a sleeved valve housing, not a fluid change.

#336 · Steering Systems

What usually causes hard spots or a catch felt at certain wheel positions in a power rack?

Answer

A scored rack, a damaged rack bearing, or a binding spool valve. The catch is felt only at the specific wheel positions where the wear or damage is located. This is different from morning sickness, which affects the whole steering range only when cold.

#337 · Steering Systems

What does a variable assist power steering system do?

Answer

It tailors the amount of boost to road speed, giving full assist for parking and reduced assist at highway speed for a firmer feel. A solenoid or actuator on the pump or gear is controlled by a module reading vehicle speed. This design balances easy parking with confident highway handling.

#338 · Steering Systems

What happens when the actuator in a variable assist power steering system fails?

Answer

It typically defaults to full assist. The usual driver complaint is steering that feels too light at highway speed. Diagnosis starts with a scan tool check of the actuator circuit.

#339 · Steering Systems

Why should you never mix incompatible power steering fluids?

Answer

Different manufacturers specify different fluid types, and mixing the wrong ones swells the internal rubber seals. Swollen seals then leak or fail early. Always confirm the exact fluid specification in the service information before topping up or refilling.

#340 · Steering Systems

Why must hoses be inspected as part of power steering diagnosis?

Answer

A collapsing return hose restricts flow and reduces assist, while a leaking pressure hose near hot exhaust is a fire hazard. Hoses should be checked for leaks, chafing, and internal deterioration. A hose problem can mimic a pump or gear fault if it is overlooked.

#341 · Steering Systems

Where is the electric motor located in electric power steering (EPS)?

Answer

It can be mounted on the steering column, on the pinion, or concentric with the rack itself. Column mount units suit lighter vehicles and keep the motor inside the cabin. Pinion mount and rack mount units handle the higher steering loads of heavier vehicles.

#342 · Steering Systems

What does the torque sensor in an EPS system measure?

Answer

It usually reads the twist of a torsion bar on the input shaft, sensing how hard and in which direction the driver is turning the wheel. This signal tells the control module how much assist to add. The module then scales the motor's output by both vehicle speed and steering rate.

#343 · Steering Systems

Name three advantages of electric power steering (EPS) over hydraulic power steering.

Answer

There is no fluid, no hoses, and no pump drag on the engine, which saves fuel. Assist is still available even with the engine off during a stall. The system also allows software control of steering feel and eliminates seasonal fluid viscosity changes.

#344 · Steering Systems

What is the main trade-off of electric power steering (EPS) compared to hydraulic systems?

Answer

EPS depends heavily on good battery voltage and clean grounds, so low system voltage causes reduced or intermittent assist. Repeated full lock parking maneuvers can also make the module reduce assist temporarily to protect the motor from overheating. There is nothing to leak, but there are new electrical failure points.

#345 · Steering Systems

When does an EPS system typically need a calibration or reset?

Answer

After a wheel alignment, after the battery is disconnected, or after the column or gear has been replaced. The calibration usually includes a steering angle sensor centering routine and a torque sensor zero point learn. It is done with a scan tool or a specified drive cycle.

#346 · Steering Systems

What can happen if EPS calibration is skipped after an alignment or battery disconnect?

Answer

The vehicle can develop a pull, an off-center steering wheel feel, or stability control faults. This happens because the steering angle sensor also feeds data to the ABS and stability control systems. A missed calibration can therefore cause problems well beyond steering feel.

#347 · Steering Systems

What happens when the control module detects an EPS fault?

Answer

It sets a diagnostic trouble code, lights the EPS warning lamp, and usually shuts off electric assist entirely. The vehicle can still be steered manually, but effort becomes very high. This is a safety fallback rather than a total loss of steering.

#348 · Steering Systems

How can some EPS systems compensate for a slight steady steering pull using a scan tool?

Answer

A scan tool can command a small constant motor torque to offset a mild pull caused by road crown or minor alignment tolerance. This is a software compensation, not a mechanical fix. It should only be used for a small, steady pull, not to mask an actual alignment or component problem.

#349 · Steering Systems

What EPS symptom points to a faulty torque sensor?

Answer

A dead spot in steering feel or noticeably uneven assist as the wheel is turned. This usually means the torque sensor needs recalibration or replacement. The torque sensor is the component that tells the module how hard and which way the driver is steering.

#350 · Steering Systems

What do four wheel steering systems do at low speed?

Answer

They steer the rear wheels a few degrees opposite to the front wheels. This shrinks the vehicle's turning circle, making tight parking and low speed maneuvers easier. At higher speed the behavior changes to improve stability instead.

#351 · Steering Systems

What do four wheel steering systems do at higher vehicle speed?

Answer

They steer the rear wheels in the same direction as the front wheels rather than opposite. This improves stability during lane changes and highway maneuvers. The system switches its behavior based on vehicle speed.

#352 · Steering Systems

What controls the rear wheel angle in a modern four wheel steering system?

Answer

An electric rear actuator, controlled by a module that reads vehicle speed and front steering angle. The module decides both the direction and the amount of rear steer needed. This replaces older purely mechanical four wheel steering linkages.

#353 · Steering Systems

Why must a four wheel steering system be centered or placed in its service mode before a rear alignment?

Answer

Rear toe and the actuator's center position both affect the alignment reading. If the actuator is not centered first, the rear toe measurement will not reflect the vehicle's true resting geometry. Skipping this step can lead to an incorrect alignment.

#354 · Steering Systems

What key service points affect rear alignment on a four wheel steering vehicle?

Answer

Rear toe setting and the rear actuator's center position both matter and must be verified together. The manufacturer's service procedure usually requires putting the system into a service mode before taking readings. Ignoring the actuator's position can produce a misleading toe reading.

#355 · Steering Systems

A vehicle has hard steering in both directions with a hydraulic power steering system. What should you check first?

Answer

Check fluid level, belt condition, pump output, tire pressure, and any binding at the ball joints or strut bearings. A weak pump or low fluid are common simple causes. If effort stays high even with the engine off, the problem is more likely a mechanical bind than a hydraulic loss.

#356 · Steering Systems

A vehicle steers hard in only one direction. What does this point to?

Answer

An internal steering gear problem, usually a leaking rack piston seal or a spool valve fault feeding pressure unevenly. The pump itself cannot know which direction you are turning, so a directional problem points inside the gear rather than at the pump. This is a useful diagnostic shortcut on the exam.

#357 · Steering Systems

A driver complains of steering looseness and wander. What is the general diagnostic order?

Answer

Work outward from the wheel: check the intermediate shaft joints first, then the gear adjustment or rack mounts, then every linkage socket, then the wheel bearings. Looseness and wander point outward from where the driver feels it. Working systematically avoids missing a wear point.

#358 · Steering Systems

What causes poor returnability, where the steering wheel does not come back after a turn?

Answer

An overtight gear or rack yoke adjustment, binding upper strut mounts or ball joints, or seized linkage sockets. An alignment cause such as insufficient caster can also produce this symptom. Poor returnability has both mechanical and alignment causes, so both must be checked.

#359 · Steering Systems

What does a whine or growl on turns that changes with engine speed usually indicate?

Answer

Power steering pump aeration or internal pump wear. Because it tracks engine rpm, it points back to the hydraulic pump rather than the linkage or gear. This is different from a clunk or a pop noise, which have mechanical causes.

#360 · Steering Systems

What does a clunk on direction change during steering usually indicate?

Answer

A loose rack mount, a worn steering coupler, or a worn linkage joint. This noise is mechanical and happens specifically when the direction of steering effort reverses. It should not be confused with pump whine, which tracks engine speed instead.

#361 · Steering Systems

What does a pop or creak heard only at full steering lock suggest?

Answer

A worn strut bearing or a bellows boot that is contacting another part at full travel. This noise appears specifically at the extreme of steering travel rather than throughout the turn. It is a different cause than a clunk on direction reversal.

#362 · Steering Systems

The steering wheel is off center right after tie rod work, but total toe measures in spec. What is the cause and fix?

Answer

The tie rods were adjusted unequally between the two sides, so the split of toe between left and right is wrong even though the total is correct. The fix is to adjust both tie rods in equal and opposite amounts. Adjust until the wheel is centered while toe stays within specification.

#363 · Steering Systems

Why is an off-center steering wheel after an alignment considered a serious comeback?

Answer

It is the single most common comeback after an alignment job. The vehicle can drive perfectly straight while the wheel itself sits crooked, which customers notice immediately. Always confirm wheel centering as part of the final road test, not just straight tracking.

#364 · Steering Systems

What should a technician check with the engine off when diagnosing hard steering in both directions?

Answer

Check whether steering effort stays high even without hydraulic assist running. If it does, the cause is more likely a mechanical bind, such as tight ball joints, strut bearings, or low tire pressure, rather than a hydraulic power loss. This step separates hydraulic problems from mechanical ones.

#365 · Steering Systems

What is the pump versus gear diagnostic logic used in power steering pressure testing?

Answer

If the restricted test shows low maximum pressure, the pump or its relief valve is at fault. If maximum pressure is good but effort is high while turning against the gear with the valve open, the leak is inside the gear itself. This comparison isolates the fault to one component without guessing.

#366 · Steering Systems

Why does a torn inner tie rod boot lead to steering noise or looseness over time on a power rack?

Answer

A torn boot lets water and grit into the inner socket, wearing it out quickly and eventually causing lost motion between the rack and tie rod. This can produce clunks or a rattly feel over bumps as wear develops. Boot inspection is part of every steering system check for this reason.

#367 · Steering Systems

What effect does low system voltage have on an EPS equipped vehicle?

Answer

It can cause reduced or intermittent power steering assist, since the electric motor and control module both depend on stable battery voltage. Poor grounds have the same effect as low voltage. This is a distinct failure mode that hydraulic systems do not have.

#368 · Steering Systems

Why do repeated full lock parking maneuvers stress an EPS motor?

Answer

Full lock steering demands maximum motor torque and current for an extended time, generating heat in the motor. The control module can respond by reducing assist temporarily to protect the motor from damage. This is the electric equivalent of the hydraulic system's overheating risk at full lock.

#369 · Steering Systems

What noise complaint results from a restricted suction hose or a clogged reservoir screen on a hydraulic power steering pump?

Answer

Cavitation, which produces a rattling growl similar to aerated fluid. The pump is being starved for fluid on the suction side, causing vapor bubbles to form and collapse. Checking the suction hose and reservoir screen is the fix, not just adding fluid.

#370 · Steering Systems

Why should a technician confirm the exact power steering fluid specification before topping up a system?

Answer

Different vehicles require different fluids, including dedicated power steering fluid, automatic transmission fluid, or a specific synthetic blend. Using the wrong fluid can swell internal seals and cause leaks that look like a hardware failure. Always check service information rather than assuming one fluid fits all vehicles.

#371 · Steering Systems

What is the difference between aeration and cavitation in a power steering pump?

Answer

Aeration is air mixed into the fluid, usually from a low fluid level or a suction side air leak, and it shows up as foamy fluid and a whine. Cavitation is vapor bubbles forming and collapsing because the pump is starved of fluid, producing a similar rattling growl. Both point to a suction side problem but are technically different phenomena.

#372 · Steering Systems

What does a scan tool check first when a variable assist actuator fault is suspected?

Answer

The actuator circuit, since a failed actuator typically defaults the system to full assist. The complaint in that case is steering that feels too light at highway speed. Confirming the circuit fault first avoids unnecessary teardown of the pump or gear.

#373 · Steering Systems

Why is EPS considered better for fuel economy than hydraulic power steering?

Answer

A hydraulic pump is driven constantly by the engine through a belt, creating drag even when no steering assist is needed. An EPS motor only draws power when the driver is actually steering. Removing constant pump drag reduces parasitic engine load and improves fuel economy.

#374 · Suspension Systems and Frames

What is sprung weight?

Answer

Sprung weight is everything the springs carry, including the body, frame, engine, transmission, and passengers. It sits above the springs and is isolated from road shocks by them. This is one of the two key weight terms used constantly on the 310E exam.

#375 · Suspension Systems and Frames

What is unsprung weight?

Answer

Unsprung weight is everything that moves with the wheel, such as the wheel and tire, brake rotor and caliper, hub, and roughly half the weight of the control arms and springs. It is not supported by the springs. Lower unsprung weight lets the wheel follow the road more easily.

#376 · Suspension Systems and Frames

Why do heavy solid axles ride more harshly than lighter independent designs?

Answer

A heavy solid axle adds a lot of unsprung weight, which the spring and shock must control. With more mass moving with the wheel, the suspension has a harder time keeping the tire planted over bumps. Lighter independent suspensions have less unsprung weight, so they absorb shocks more smoothly.

#377 · Suspension Systems and Frames

What is jounce?

Answer

Jounce, also called compression, is upward wheel travel toward the body, such as when a tire hits a bump. It is one of the two basic wheel movement terms tested on the exam. The opposite movement is called rebound.

#378 · Suspension Systems and Frames

What is rebound in suspension terms?

Answer

Rebound, also called extension, is downward wheel travel away from the body as the spring pushes the wheel back after a bump. It is controlled mainly by the shock absorber or strut. Rebound is the opposite of jounce.

#379 · Suspension Systems and Frames

What does a jounce bumper (bump stop) do?

Answer

A jounce bumper limits maximum compression so the suspension never bottoms out metal to metal. It protects components during hard impacts or heavy loading. Rebound travel, by contrast, is limited by the shock absorber or strut, not the bump stop.

#380 · Suspension Systems and Frames

What defines a short-long arm (SLA) suspension?

Answer

An SLA, or double wishbone, design uses an upper control arm that is shorter than the lower control arm. This unequal arm length is deliberate and controls how the tire tilts during suspension travel. SLA suspensions carry the spring on one of the arms and use a conventional shock absorber.

#381 • Suspension Systems and Frames

Why is the upper arm shorter than the lower arm on an SLA suspension?

Answer

During jounce, the short upper arm swings through a tighter arc than the longer lower arm. This tilts the top of the tire inward slightly, keeping the tread close to its original track width. The result is reduced scrub and less tire wear during suspension travel.

#382 • Suspension Systems and Frames

What is a MacPherson strut?

Answer

A MacPherson strut combines the shock absorber, spring seat, and upper steering pivot into one assembly, with the upper mount bolted to a tower in the unibody. Only a single lower control arm is needed because the strut replaces the upper arm and upper ball joint. It is compact, light, and inexpensive, which is why it dominates front-wheel-drive cars.

#383 • Suspension Systems and Frames

Where does the steering axis run on a MacPherson strut front end?

Answer

On a strut front end, the steering axis runs through the upper strut mount bearing and the lower ball joint. This is different from an SLA design, which uses two ball joints to define the steering axis. A bent strut on this design affects alignment angles directly.

#384 • Suspension Systems and Frames

How does a torsion bar suspension work?

Answer

A torsion bar suspension uses a long steel bar that twists in place of a coil spring. One end anchors to the lower control arm and the other anchors to the frame through an adjustable anchor arm. As the wheel moves up, the bar twists and resists, acting just like an unwound coil spring.

#385 • Suspension Systems and Frames

How is ride height adjusted on a torsion bar suspension?

Answer

Ride height on a torsion bar suspension is adjusted by turning the adjuster bolt at the anchor point on the frame. This is one of the few suspensions where ride height is adjustable by design. Turning the bolt changes how much the bar is preloaded, raising or lowering the vehicle.

#386 • Suspension Systems and Frames

What is the twin I-beam front suspension?

Answer

Twin I-beam is a front suspension used on Ford light trucks where each front wheel mounts on its own long I-beam axle pivoting near the opposite frame rail, with a radius arm controlling fore and aft movement. It combines beam axle strength with some independent wheel movement. Its long swing arcs cause noticeable camber change through jounce and rebound, leading to tire wear complaints.

#387 · Suspension Systems and Frames

How does a solid rear axle on leaf springs locate the axle?

Answer

On a solid (live) rear axle with leaf springs, the leaf springs both support the load and locate the axle, so no separate control arms are needed. This is the simplest rear suspension arrangement. It keeps both rear wheels parallel so rear camber never changes with body roll.

#388 · Suspension Systems and Frames

Why does a solid rear axle on coil springs need control arms?

Answer

A coil spring can only push and pull straight up and down, so it cannot locate the axle fore, aft, or sideways the way a leaf spring can. A solid axle on coil springs therefore needs control arms, usually two lower trailing arms and two upper arms or a track bar, to hold the axle in position. Without them the axle would shift under acceleration and braking forces.

#389 · Suspension Systems and Frames

What is a trailing arm rear suspension?

Answer

A trailing arm mounts each rear wheel on an arm that pivots at its front end, with the pivot axis square to the vehicle centerline. Because the axis is square, camber and toe stay constant through wheel travel. This makes trailing arm suspensions predictable for alignment purposes.

#390 · Suspension Systems and Frames

What is a semi-trailing arm rear suspension?

Answer

A semi-trailing arm is similar to a trailing arm but pivots on an axis angled relative to the vehicle centerline. This angled pivot produces some camber and toe change as the wheel moves through jounce and rebound. It is a common source of exam questions about geometry change during suspension travel.

#391 • Suspension Systems and Frames

What is a twist beam (torsion beam) rear axle?

Answer

A twist beam rear axle is a pair of trailing arms joined by a flexible crossmember, commonly used on front-wheel-drive cars. The crossmember flexes to allow some independent wheel movement and also acts partly like a stabilizer bar. It is simpler and cheaper than a full independent rear suspension.

#392 • Suspension Systems and Frames

What is a multi-link rear suspension?

Answer

Multi-link rear suspension uses three, four, or five separate links per wheel, with each link controlling one direction of wheel movement. This lets designers tune camber and toe behaviour very precisely. It is standard on most modern sedans and SUVs because of the ride and handling benefits it provides.

#393 • Suspension Systems and Frames

What is the main trade-off of independent rear suspension designs?

Answer

Independent rear designs reduce unsprung weight and let each wheel react to bumps separately, which improves ride quality and grip. The trade-off is more bushings, links, and alignment adjustment points that must be inspected. Worn bushings on a multi-link car allowing toe or camber drift are a prime suspect for inner edge rear tire wear.

#394 • Suspension Systems and Frames

What is spring rate?

Answer

Spring rate is the amount of force needed to compress a spring a given distance, expressed in newtons per millimetre or pounds per inch. It describes how stiff or soft a spring is. A higher spring rate means the spring resists compression more strongly.

#395 · Suspension Systems and Frames

What is a linear rate spring?

Answer

A linear rate spring compresses the same amount for each equal increase in load. Its stiffness stays constant throughout its travel. This is the simplest type of spring behaviour to describe and calculate.

#396 · Suspension Systems and Frames

What is a progressive (variable) rate spring?

Answer

A progressive rate spring gets stiffer as it compresses further, achieved with unequal coil spacing, a tapered wire diameter, or a tapered coil shape. It rides softly over small bumps but resists bottoming out under heavy loads. This gives good comfort and load capacity from a single spring design.

#397 · Suspension Systems and Frames

What is the rule for replacing coil springs?

Answer

Coil springs must be replaced in pairs on the same axle when weak or broken. Any sag beyond the ride height specification means the spring itself has failed, not the shock absorber. Replacing only one spring on an axle can cause uneven ride height and handling.

#398 · Suspension Systems and Frames

A vehicle sits low at ride height. Why is replacing the shock absorber the wrong fix?

Answer

Shock absorbers control spring motion and dampen bouncing, but they do not support the vehicle's weight. A low ride height means the spring has sagged and lost its ability to hold the vehicle at the correct height. The correct fix is spring replacement, not shock replacement.

#399 · Suspension Systems and Frames

What does the centre bolt do on a leaf spring, and what happens if it breaks?

Answer

The centre bolt clamps the stack of leaf spring blades together at their centre and keeps the axle centered on the spring. If the centre bolt breaks, the axle can shift on the spring, changing the wheelbase on that side. This pulls the thrust angle off, which is a favourite exam scenario.

#400 · Suspension Systems and Frames

What do rebound clips do on a leaf spring?

Answer

Rebound clips hold the individual leaves of a leaf spring in line with each other as the spring flexes. They prevent the leaves from separating or shifting sideways during rebound travel. Plastic or zinc pads are often placed between leaves as well to reduce squeaks.

#401 • Suspension Systems and Frames

How does an air spring work?

Answer

An air spring uses a reinforced rubber bag inflated by an onboard compressor instead of a steel coil. Because the air pressure can be varied, air suspension can maintain a level ride height regardless of vehicle load. Height sensors at each corner tell the control module when to add or vent air.

#402 • Suspension Systems and Frames

Why must some air suspension systems be switched off before lifting the vehicle?

Answer

When the vehicle is lifted, the suspension extends fully and the height sensors read a low ride height. If the system is left on, the compressor tries to fill the fully extended bags to reach the target height. This can overextend or damage the air springs, so the system must be disabled first.

#403 · Suspension Systems and Frames

Why are torsion bars directional (left side versus right side)?

Answer

A torsion bar is stress-relieved for twisting in only one direction during manufacturing. A bar made for the left side must never be installed on the right, because it is not built to resist twist the opposite way. Installing it on the wrong side can cause premature failure.

#404 · Suspension Systems and Frames

What does a shock absorber actually do?

Answer

A shock absorber controls the oscillation of the spring after it is compressed by a bump. Without damping, a compressed spring would bounce the body several times before settling down. The shock forces hydraulic fluid through small orifices and valves as its piston moves, converting motion into heat.

#405 • Suspension Systems and Frames

Why are most modern shocks gas charged?

Answer

Gas charging prevents the hydraulic fluid inside from aerating or foaming during rapid, repeated strokes, which keeps the damping force consistent over rough roads. Twin tube shocks typically use low-pressure nitrogen in a separate reservoir tube, while mono tube shocks use a floating piston pressurized with high-pressure gas.

#406 • Suspension Systems and Frames

What is a twin tube shock absorber?

Answer

A twin tube shock has an inner working tube and an outer reservoir tube, separated by a base valve. It is relatively inexpensive and can tolerate light dents in the outer tube without failing. This design has been used in shock absorbers for many decades.

#407 · Suspension Systems and Frames

What is a mono tube shock absorber, and what is its main weakness?

Answer

A mono tube shock has a single tube with a floating piston separating high-pressure gas from the fluid inside. It dissipates heat better, can be mounted inverted, and responds faster than a twin tube design. Its main weakness is that a dent in the single tube ruins it, since the piston rides directly on that surface.

#408 · Suspension Systems and Frames

What is a strut, as distinct from a plain shock absorber?

Answer

A strut is a shock absorber built into a structural housing that also locates the wheel and carries the coil spring. A plain shock absorber only dampens motion and does not locate the wheel or carry spring load. Many struts use a replaceable cartridge, though most modern ones are replaced as a complete unit or loaded assembly.

#409 · Suspension Systems and Frames

How is the bounce test performed on a shock or strut?

Answer

The technician pushes down firmly on each corner of the vehicle and then releases it. A good shock or strut lets the body rise, settle, and stop after no more than one full oscillation, about one and a half bounces. Continued bouncing after that indicates weak damping.

#410 · Suspension Systems and Frames

How do you tell acceptable shock seepage from a failed shock seal during a visual inspection?

Answer

A light film or misting of oil on the shock body is considered acceptable and does not require replacement. Fluid that is actually running down the housing or dripping off the unit means the seal has failed. A failed shock or strut must be replaced when this is found.

#411 • Suspension Systems and Frames

What is the rule for replacing shocks or struts?

Answer

Shocks and struts, like springs, are replaced in axle pairs rather than one at a time. Replacing only one side can cause uneven damping and inconsistent handling between the two wheels. This mirrors the same in-pairs rule that applies to coil spring replacement.

#412 • Suspension Systems and Frames

A customer complains of cupped or scalloped tire wear. What does that pattern usually point to?

Answer

Cupped or scalloped tire wear points to worn shock absorbers or struts that are letting the tire hop over bumps instead of staying planted. When damping is weak, the tire loses and regains contact with the road repeatedly, wearing an uneven scalloped pattern. This is a classic diagnostic clue tested on the exam.

#413 • Suspension Systems and Frames

What does excessive nosedive under braking usually indicate?

Answer

Excessive nosedive under braking points to weak front shocks or struts that can no longer control the forward weight transfer. As braking forces load the front suspension, weak dampers allow it to compress too much and too quickly. This is a common ride complaint linked directly to damper condition.

#414 • Suspension Systems and Frames

What is the difference between a load-carrying ball joint and a follower (friction) ball joint?

Answer

The load-carrying ball joint is mounted on the control arm that carries the spring, so it supports vehicle weight and wears fastest. The follower, or friction, joint on the other arm simply maintains geometry and carries little load. Identifying which joint carries the load is a core exam skill, since it determines how the joint must be measured.

#415 · Suspension Systems and Frames

On an SLA suspension with the spring on the lower arm, which ball joint is the load carrier? What about a MacPherson strut?

Answer

On this SLA design, the lower ball joint is the load carrier because the spring load passes through the lower arm. On a MacPherson strut, the strut itself carries the spring load, so the single lower ball joint on that design is a follower type instead. This difference changes how each joint must be unloaded for measurement.

#416 · Suspension Systems and Frames

How do you measure a load-carrying ball joint for wear?

Answer

The spring load must first be removed from the joint before measuring. If the spring sits on the lower arm, support the lower control arm so the joint hangs free; if it acts on the upper arm, support the frame so the lower arm hangs instead. Then measure axial (vertical) and radial (horizontal) movement with a dial indicator while prying the wheel, and compare the readings to the manufacturer's specification.

#417 · Suspension Systems and Frames

How does a built-in wear indicator ball joint show that it is worn out?

Answer

This design uses a grease fitting boss that protrudes from the housing when the joint is new. As the joint wears, the shoulder recedes until it sits flush with or below the housing, showing the joint is worn out. Unlike conventional joints, indicator joints are checked loaded, with the vehicle sitting on its wheels.

#418 · Suspension Systems and Frames

What do control arm bushings do, and what happens when they fail?

Answer

Control arm bushings are rubber or hydraulic mounts that isolate road noise and allow the arm to pivot in a controlled way. Worn, cracked, oil-soaked, or torn bushings let the arm shift out of position, causing clunks, wander, and alignment drift. They can also cause brake pull as the wheel moves rearward under braking.

#419 · Suspension Systems and Frames

Why must control arm pivot bolts only be tightened at normal ride height?

Answer

Rubber bushings are designed to sit in a relaxed, unstressed position at normal ride height. Torquing the pivot bolts while the suspension is at full droop pre-winds the rubber bushing before it is even in service. This pre-loading causes the bushing to fail early once the vehicle is back on its wheels.

#420 · Suspension Systems and Frames

What is a stabilizer bar (sway bar) and what does it do?

Answer

It is a torsion spring mounted across the chassis and connected to each lower control arm or strut through end links. When both wheels move together it just rotates and does nothing. When one wheel rises relative to the other, such as in a corner, the bar twists and resists body roll.

#421 · Suspension Systems and Frames

What happens when both front wheels hit a bump together on a car with a stabilizer bar?

Answer

The stabilizer bar simply rotates freely in its mounting bushings and adds no resistance. It only starts twisting and resisting when one wheel moves differently than the other, such as during cornering.

#422 · Suspension Systems and Frames

What are the symptoms of a worn or disconnected stabilizer bar link?

Answer

The customer usually reports a rattle or clunk over bumps along with increased body lean in corners. A bad link does not change ride height and does not affect any wheel alignment angle, which helps rule it out from other complaints.

#423 · Suspension Systems and Frames

Does a broken stabilizer bar link change wheel alignment?

Answer

No. A stabilizer bar and its links only control body roll during cornering, they do not support weight or locate the wheel. A worn link causes clunking and extra lean, but camber, caster, and toe stay unaffected.

#424 · Suspension Systems and Frames

How do adaptive (active) dampers change firmness in an electronic ride control system?

Answer

They use either a solenoid valve that changes the size of the fluid orifice or magnetorheological fluid whose viscosity changes when exposed to a magnetic field. Both methods let the control module make the damper firmer or softer within a fraction of a second.

#425 · Suspension Systems and Frames

What inputs does an electronic ride control module use to adjust damper firmness?

Answer

It reads wheel position sensors, steering angle, vehicle speed, and body accelerometers many times per second. Based on those inputs it firms up the dampers for cornering and braking and softens them for normal cruising.

#426 · Suspension Systems and Frames

How does an air suspension system provide automatic load levelling?

Answer

A height sensor at each corner, usually a rotary lever sensor or a Hall effect sensor, reports the gap between the body and the suspension to the control module. The module then runs the compressor to add air or opens a vent valve to remove it, holding the target ride height regardless of load.

#427 · Suspension Systems and Frames

A vehicle with air suspension sags overnight after sitting. What does this point to?

Answer

This points to a leak somewhere in the system, most likely a leaking air bag, a cracked line, or a bad valve. The compressor cannot hold pressure against a leak while the vehicle sits with the engine off.

#428 · Suspension Systems and Frames

An air suspension compressor runs almost constantly. What does that indicate and what is the risk?

Answer

A compressor that runs constantly is fighting a leak somewhere in the bags, lines, or valves rather than doing a normal occasional top-up. Left uncorrected, the constant duty cycle will overheat and burn out the compressor.

#429 · Suspension Systems and Frames

Why must some air suspension systems be switched off before lifting the vehicle on a hoist?

Answer

With the system left on, the wheels hang at full droop and the control module reads this as very low ride height. The compressor then tries to fill the fully extended air bags, which can overextend and damage them.

#430 · Suspension Systems and Frames

What must be done after replacing an air suspension height sensor or its linkage?

Answer

Most systems require a height calibration using a scan tool so the module learns the new sensor reference points. Since this changes ride height, a wheel alignment check should follow as well.

#431 · Suspension Systems and Frames

What is a ladder frame and where is it commonly used?

Answer

A ladder frame is built from two full length steel rails joined by crossmembers, with the body bolted on top through rubber mounts. It is strong and simple, and it is the standard structure on pickup trucks and many truck based SUVs.

#432 · Suspension Systems and Frames

What is a perimeter frame and how does it differ from a ladder frame?

Answer

A perimeter frame routes its rails outward around the passenger compartment instead of running straight underneath it. This gives a lower floor and better side impact protection, and it was common on older full size cars.

#433 · Suspension Systems and Frames

What is unibody (unitized) construction?

Answer

There is no separate bolt on frame. Stamped panels, rails, and pillars are welded together into a single structure, so the body itself carries all of the vehicle's loads. Nearly all modern cars and crossovers are built this way.

#434 · Suspension Systems and Frames

What is a subframe (cradle) on a unibody vehicle?

Answer

It is a bolted in structure under the engine that supports the powertrain, the lower control arms, and often the steering rack. It is a separate piece bolted to the unibody rather than part of the welded structure.

#435 · Suspension Systems and Frames

A unibody vehicle has alignment readings that are strangely offset from one side to the other on the front axle. What should the technician check?

Answer

A shifted subframe should be checked, since it changes alignment angles on both front wheels at the same time. Subframe position should always be verified whenever front alignment numbers look mismatched side to side.

#436 · Suspension Systems and Frames

What is the datum line (datum plane) used in frame measurement?

Answer

It is an imaginary horizontal plane below the vehicle that vertical heights to specific control points are measured from. It comes from the manufacturer's dimension chart and gives a consistent reference for every measurement.

#437 · Suspension Systems and Frames

What is the centerline in frame and underbody measurement?

Answer

It is an imaginary line running lengthwise through the exact centre of the vehicle. Width measurements from this line to control points on each side should be equal if the underbody structure is straight and undamaged.

#438 · Suspension Systems and Frames

What is X-checking (cross measuring) a frame or unibody?

Answer

It means measuring diagonally from a control point on one side to the mirror image point on the opposite side, then measuring the opposite diagonal. Equal diagonals within about 3 mm mean that section is square, while a larger difference means damage.

#439 · Suspension Systems and Frames

What tool lets a technician take point to point frame measurements when parts block a straight tape path?

Answer

A tram gauge with adjustable pointers can reach around obstructions and still take an accurate point to point measurement. This makes it useful for checking control points that a tape measure cannot reach in a straight line.

#440 · Suspension Systems and Frames

What is diamond damage on a frame?

Answer

Diamond damage happens when one frame rail is pushed rearward relative to the other, turning the frame's rectangle shape into a parallelogram. It shows up as unequal X-check diagonals and is usually caused by a hard corner impact.

#441 · Suspension Systems and Frames

What is sag damage on a frame?

Answer

Sag means a rail has bowed downward, usually in the section between the wheels. It is caused by a collision or by chronic overloading, and it shows up as a low datum height reading at the affected point.

#442 · Suspension Systems and Frames

What is sway (sidesway) damage on a frame?

Answer

Sway means the frame has bent sideways so the front or rear section no longer tracks along the centerline. It is typically caused by a side impact, and the vehicle will dog-track down the road as a result.

#443 · Suspension Systems and Frames

What is twist damage on a frame?

Answer

Twist means one rail rides higher than the other, so the frame is no longer flat across its width. The vehicle leans to one side and datum height readings disagree when compared diagonally.

#444 · Suspension Systems and Frames

What is mash (collapse) damage on a frame?

Answer

Mash means a section of rail has been shortened by a frontal or rear impact. Length measurements between control points come up short, and visible buckles usually appear in the rail at the damaged section.

#445 • Suspension Systems and Frames

Why does real collision damage often show more than one of the five classic frame damage types at once?

Answer

A hard enough impact can push, bend, and shorten the structure in more than one way at the same time. This is why a technician should measure the whole structure with the dimension chart rather than assuming only the obviously damaged area is affected.

#446 • Suspension Systems and Frames

Control point measurements on a vehicle come back out of specification. What is the correct next step?

Answer

The vehicle must go to a collision repair facility with a frame straightening system before any alignment is attempted. No amount of alignment adjustment can correct a bent frame or unibody structure.

#447 · Suspension Systems and Frames

What is ride height (curb height or trim height)?

Answer

It is the manufacturer's specified distance from defined chassis points to the ground, or between certain suspension points, measured on a level surface. It is measured at curb weight, meaning full fuel, correct tire pressures, and no passengers or cargo, unless the spec says otherwise.

#448 · Suspension Systems and Frames

How much side to side difference in ride height do most manufacturers allow?

Answer

Most manufacturers allow only a small side to side difference, commonly in the range of about 10 to 15 mm. Anything beyond that range signals a problem such as a weak spring or sagging suspension on one corner.

#449 · Suspension Systems and Frames

Why is ride height checked before every wheel alignment?

Answer

Suspension geometry is designed around the correct ride height, so a sagging vehicle has camber, caster, and toe values shifted away from their designed targets. Aligning a vehicle before fixing ride height only hides the symptom on the machine's readout, not in the actual geometry.

#450 · Suspension Systems and Frames

A vehicle has correct alignment readings on the machine but the customer still complains of tire wear or poor handling. What is a likely cause the exam expects?

Answer

Low ride height is a frequent intended answer in this scenario. The alignment numbers can look fine at rest while the suspension geometry through jounce and rebound remains wrong because ride height was never corrected first.

#451 • Suspension Systems and Frames

What is the correct service sequence when a vehicle has both low ride height and worn alignment angles?

Answer

First measure ride height, then replace weak springs or adjust torsion bars as needed to bring height back to specification. Only after ride height is corrected should the vehicle be aligned.

#452 • Suspension Systems and Frames

A vibration changes with vehicle speed but not with engine speed. What does that point to?

Answer

This points to the wheels, tires, hubs, or brakes rather than the powertrain. Since the vibration tracks road speed instead of engine RPM, an out of balance wheel and tire assembly is a common cause.

#453 · Suspension Systems and Frames

At what speed range does wheel and tire imbalance typically become noticeable, and how is front versus rear imbalance felt?

Answer

It typically shows up between about 80 and 110 km/h. A front imbalance is usually felt in the steering wheel, while a rear imbalance is usually felt in the seat instead.

#454 · Suspension Systems and Frames

A vibration is tied directly to engine speed rather than road speed. What does this suggest?

Answer

This points to the powertrain mounts or an accessory rather than the suspension or tires. Because the vibration changes with RPM instead of vehicle speed, the suspension system can largely be ruled out.

#455 · Suspension Systems and Frames

A shake is felt only while braking and disappears otherwise. What does this suggest?

Answer

This points to brake rotor thickness variation rather than a suspension component. Since the shake only occurs during the braking event itself, the suspension can generally be ruled out as the cause.

#456 · Suspension Systems and Frames

What does a clunk noise over bumps usually suggest?

Answer

A clunk over bumps suggests worn stabilizer links, worn control arm bushings, or a loose strut mount. All three allow small amounts of unwanted movement that produce an audible knock when the suspension compresses or rebounds.

#457 · Suspension Systems and Frames

What does a creak or groan during slow turns or while rocking the vehicle suggest?

Answer

This suggests dry or worn ball joints. The lack of lubrication or excessive wear in the joint causes friction that produces the creaking or groaning sound during slow steering movement.

#458 · Suspension Systems and Frames

What does a snap or pop while turning at low speed suggest?

Answer

This suggests a worn upper strut mount bearing, or on a front wheel drive vehicle it can suggest a worn outer CV joint. Both involve a component that binds and then releases suddenly as the wheel turns.

#459 • Suspension Systems and Frames

What does a single clunk when shifting between drive and reverse suggest?

Answer

This suggests worn control arm bushings or worn engine cradle mounts that let components shift slightly under the sudden change in torque direction. The clunk happens once at the moment of the shift rather than continuously.

#460 • Suspension Systems and Frames

Why should a technician always confirm a suspected worn part before replacing it during noise diagnosis?

Answer

Noise diagnosis by guesswork often replaces good parts while leaving the actual cause untouched, so the customer's complaint remains unresolved. A physical inspection, pry bar check, or dial indicator measurement should always confirm the suspected part first.

#461 • Suspension Systems and Frames

What is sprung weight?

Answer

Sprung weight is everything the springs carry, including the body, frame, engine, transmission, and passengers. It is separate from unsprung weight, which moves with the wheel instead of resting on the springs.

#462 • Suspension Systems and Frames

What is unsprung weight, and why does lowering it improve ride quality?

Answer

Unsprung weight is everything that moves with the wheel, including the wheel and tire, brake rotor and caliper, hub, and roughly half the weight of the springs, control arms, and axle shafts. Lower unsprung weight lets the wheel follow the road more easily because the spring and shock have less mass to control.

#463 • Suspension Systems and Frames

What is jounce and what limits it?

Answer

Jounce (compression) is upward wheel travel toward the body, as when a tire hits a bump. Maximum compression is limited by a jounce bumper, also called a bump stop, so the suspension never bottoms metal to metal.

#464 • Suspension Systems and Frames

What is rebound and what limits it?

Answer

Rebound (extension) is downward wheel travel away from the body as the spring pushes the wheel back after a bump. Rebound travel is limited by the shock absorber or strut rather than by a separate bumper.

#465 · Suspension Systems and Frames

Why does the exam test camber and toe changes during jounce for specific suspension designs?

Answer

Different suspension geometries move the wheel through different arcs as it travels up and down, so camber and toe do not stay constant on every design. Knowing which designs change these angles during jounce, such as twin I-beam or semi-trailing arm, helps predict wear patterns and diagnose complaints.

#466 · Tires Wheels Rims and Hubs

What does radial tire construction mean?

Answer

The body ply cords run at 90 degrees to the direction of travel, straight across from bead to bead. This lets the sidewall flex independently of the tread. That keeps a larger, more stable contact patch on the road than a bias ply tire.

#467 · Tires Wheels Rims and Hubs

What is the tire bead and what does it do?

Answer

The bead is a bundle of high tensile steel wires wrapped in rubber. It anchors the tire to the rim. It also resists the force of inflation pressure trying to push the tire off the wheel.

#468 · Tires Wheels Rims and Hubs

What job do the belts do inside a tire?

Answer

Belts are normally two layers of steel cord running around the circumference under the tread. They stabilize the tread and resist punctures. They also hold the tread flat against the road.

#469 · Tires Wheels Rims and Hubs

In the size P225/60R16 97T, what does the 60 mean?

Answer

The 60 is the aspect ratio. It means the sidewall height is 60 percent of the section width. For a 225 mm section width that sidewall stands about 135 mm tall.

#470 · Tires Wheels Rims and Hubs

In P225/60R16 97T, what does the 97 tell you?

Answer

The 97 is the load index, a code number tied to a maximum load per tire. In this case it corresponds to 730 kg at the tire's maximum rated pressure. A higher load index means the tire can carry more weight.

#471 · Tires Wheels Rims and Hubs

How do you read a tire's speed rating letter?

Answer

The speed rating letter is the maximum sustained speed the tire is built for. Common ratings climb through S, T, H, V, and W as speed capability increases. A replacement tire should meet or exceed the vehicle manufacturer's original speed rating.

#472 · Tires Wheels Rims and Hubs

What is different about LT tire sizing compared to P-metric?

Answer

LT tires carry a load range letter such as C, D, or E that reflects ply strength and maximum inflation pressure. They often list two load capacities, one for single wheel use and a lower one for dual wheel use. A P-metric tire used in light truck service must have its load capacity derated by dividing by 1.10.

#473 · Tires Wheels Rims and Hubs

How do you read a tire's DOT date code, for example 3223?

Answer

The last four digits of the DOT code are the date code. The first two digits are the week of manufacture and the last two are the year. A code of 3223 means the tire was built in the 32nd week of 2023.

#474 · Tires Wheels Rims and Hubs

What does a UTQG treadwear grade of 500 actually compare?

Answer

Treadwear is a number comparing wear life to a reference tire rated 100 under controlled test conditions. A 500 rated tire should last about five times as long as that reference tire. The number is only fair to compare within one manufacturer's own tire line.

#475 · Tires Wheels Rims and Hubs

What does the UTQG traction grade actually measure?

Answer

Traction is graded AA, A, B, or C. It measures straight line wet braking performance only. It does not measure cornering grip at all.

#476 · Tires Wheels Rims and Hubs

What is the difference between the mountain snowflake symbol and the M+S marking?

Answer

The peaked mountain and snowflake symbol means the tire passed a tested snow traction performance standard. The older M+S marking is only a mud and snow tread pattern claim with no performance test behind it. In Canada, insurance discounts and Quebec's winter tire law recognize the mountain snowflake symbol, not plain M+S.

#477 · Tires Wheels Rims and Hubs

Where should you get the correct tire inflation pressure from?

Answer

Use the tire placard, normally found on the driver's door jamb. The sidewall figure is only the maximum pressure the tire can hold, not the recommended setting. Setting every tire to the sidewall maximum is a classic mistake.

#478 · Tires Wheels Rims and Hubs

Why should tire pressure be checked cold?

Answer

Cold means the vehicle has sat about three hours or been driven less than roughly 2 km. Driving heats the air inside the tire and raises the pressure reading. Bleeding a hot tire down to the cold spec leaves it underinflated once it cools.

#479 · Tires Wheels Rims and Hubs

What tread wear pattern points to overinflation?

Answer

Overinflation shrinks the contact patch so the tire rides mainly on the tread center. This causes wear concentrated across the center of the tread only. The shoulders stay relatively unworn compared to the middle.

#480 · Tires Wheels Rims and Hubs

What tread wear pattern points to underinflation?

Answer

Underinflation lets the sidewalls flex excessively and pushes more load onto the outer parts of the tread. This causes wear on both shoulders with a good center. It also builds heat and raises rolling resistance.

#481 · Tires Wheels Rims and Hubs

What causes one-shoulder-only tire wear?

Answer

One shoulder wearing more than the other points to a camber problem. Excessive negative camber wears the inside shoulder, and excessive positive camber wears the outside shoulder. A sagging spring that changed the camber angle can cause the same pattern.

#482 · Tires Wheels Rims and Hubs

What causes feathered (saw tooth) tire wear?

Answer

Feathering is caused by incorrect toe. The tire scrubs sideways slightly with every meter it rolls, wearing each rib to a sharp edge on one side and a rounded edge on the other. Toe error is the fastest tread killer of all the alignment angles.

#483 · Tires Wheels Rims and Hubs

What causes cupped or scalloped tire wear?

Answer

Cupping shows as irregular dished patches around the tread circumference. It is caused by the tire bouncing rather than by steering geometry. Suspect worn shocks or struts, a loose wheel bearing, worn ball joints, or an out of balance wheel.

#484 · Tires Wheels Rims and Hubs

What tire rotation pattern is used for non-directional tires on a front wheel drive vehicle?

Answer

The common pattern is the forward cross. The rear tires cross to the opposite front corners, and the front tires move straight back. This evens out wear since the front tires carry, drive, and steer the car.

#485 · Tires Wheels Rims and Hubs

How must directional tires be rotated?

Answer

Directional tires are marked with a rotation arrow on the sidewall and must always keep turning the same direction. They can only rotate front to rear on the same side of the vehicle. They can only move to the other side if they are dismantled and remounted.

#486 · Tires Wheels Rims and Hubs

What makes a tire puncture repairable under trade standards?

Answer

A puncture is repairable only if it lies in the crown, the tread area between the outer circumferential grooves. It must also be no larger than 6 mm (1/4 inch). Any injury to the shoulder or sidewall makes the tire non-repairable.

#487 · Tires Wheels Rims and Hubs

What is the only accepted method for repairing a tire puncture?

Answer

The only accepted repair is a combination plug and patch, or a one piece plug-patch unit. It must be installed from inside the tire after the tire is dismounted and inspected. A string plug pushed in from outside, a patch alone, or aerosol sealant do not meet trade standards.

#488 · Tires Wheels Rims and Hubs

What is the maximum pressure allowed when seating tire beads during mounting?

Answer

Never exceed 275 kPa (40 psi) when seating beads. If the beads have not seated by then, deflate the tire, relubricate the beads, reposition the tire, and try again. Exceeding this pressure risks a dangerous bead blowout.

#489 · Tires Wheels Rims and Hubs

What is the difference between static and dynamic wheel imbalance?

Answer

Static imbalance means the assembly is heavier at one point around its circumference, and it causes wheel tramp, a vertical hopping. Dynamic imbalance means weight is unevenly distributed side to side across the assembly width, causing shimmy, a side to side wobble felt in the steering wheel. Dynamic imbalance needs weights on both the inner and outer rim planes to correct.

#490 · Tires Wheels Rims and Hubs

What does road force balancing measure that ordinary spin balancing does not?

Answer

A road force balancer presses a load roller against the spinning tire to measure stiffness variation and loaded runout, called road force. A tire can be perfectly balanced yet still vibrate due to a stiff spot or excessive runout. The machine can direct match mounting to cancel much of the assembly runout.

#491 · Tires Wheels Rims and Hubs

What is the difference between radial and lateral tire runout?

Answer

Radial runout is the up and down variation, measured on the center tread rib or the bead seat area of the wheel. Lateral runout is the side to side wobble, measured on the sidewall or the vertical face of the rim. Typical service limits are about 1.5 mm for an assembled tire and wheel.

#492 · Tires Wheels Rims and Hubs

What speed range and vibration pattern points to tire and wheel imbalance or runout?

Answer

A vibration felt from about 80 to 120 km/h that changes with road speed, not engine speed, points to tire and wheel imbalance or runout. This differs from an acceleration-only vibration, which points to driveline angles or a worn joint. Always road test, then spin balance, then check runout before selling parts.

#493 · Tires Wheels Rims and Hubs

What is the difference between direct and indirect TPMS?

Answer

Direct TPMS uses a battery powered sensor inside each wheel that transmits actual measured pressure and temperature by radio. Indirect TPMS has no pressure sensors and instead uses ABS wheel speed sensors to detect a smaller rolling radius on an underinflated tire. Indirect systems cost less but cannot display actual pressures and can miss all four tires being equally low.

#494 · Tires Wheels Rims and Hubs

What should be replaced whenever a direct TPMS tire is dismantled?

Answer

The sensor's serviceable parts should be replaced with a service kit: the valve core, sealing grommet, nut, and cap. The nickel plated valve core must never be swapped for a standard brass core, since brass corrodes against the aluminum stem. The small sensor nut must be torqued to specification because it strips easily.

#495 · Tires Wheels Rims and Hubs

What is the difference between positive and negative wheel offset?

Answer

Positive offset means the mounting face sits outboard of the rim's centerline, tucking the wheel in toward the vehicle. Negative offset pushes the wheel and tire outboard, producing a deep dish look. Changing offset changes the scrub radius and the loads placed on wheel bearings and steering parts.

#496 · Tires Wheels Rims and Hubs

What is the difference between hub-centric and lug-centric wheels?

Answer

Hub-centric wheels are located by a machined centre bore that fits closely over the hub's pilot flange, so the hub centres the wheel. Lug-centric wheels are centred instead by the tapered seats of the lug nuts and must be tightened progressively with the wheel off the ground. Aftermarket wheels with oversized centre bores often need hub-centric rings to avoid vibration.

#497 · Tires Wheels Rims and Hubs

Why should you avoid oiling wheel studs before torquing lug nuts?

Answer

Lubricated threads produce far more actual clamping force at the same torque wrench reading than dry threads. This can overstretch and damage the studs. Studs should only be oiled if the manufacturer specifically calls for it.

#498 · Tires Wheels Rims and Hubs

How should lug nuts be tightened for final torque?

Answer

Snug the nuts in a star or crisscross pattern first, then lower the vehicle to the ground. Apply final torque with a calibrated torque wrench in the same star pattern, typically around 100 to 140 N.m on passenger vehicles. An impact gun alone routinely over torques, which stretches studs and can crack alloy wheels.

#499 · Tires Wheels Rims and Hubs

How do you adjust a serviceable tapered front wheel bearing?

Answer

Tighten the adjusting nut to a seating torque of roughly 20 to 30 N.m while rotating the wheel to seat the parts. Back the nut off, then retighten it finger tight or to the light value the maker specifies. Confirm end play of 0.025 to 0.13 mm with a dial indicator before installing a new cotter pin.

#500 · Tires Wheels Rims and Hubs

How do you check for play in a modern sealed wheel hub unit?

Answer

Raise the wheel and rock the tire at the 12 and 6 o'clock positions. To separate bearing play from steering linkage play, also rock at 3 and 9 o'clock or have a helper apply the brake pedal. Allowable play in a sealed hub is nearly zero, generally under about 0.05 mm.